Chapter 1: Introduction

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April 14, 2026

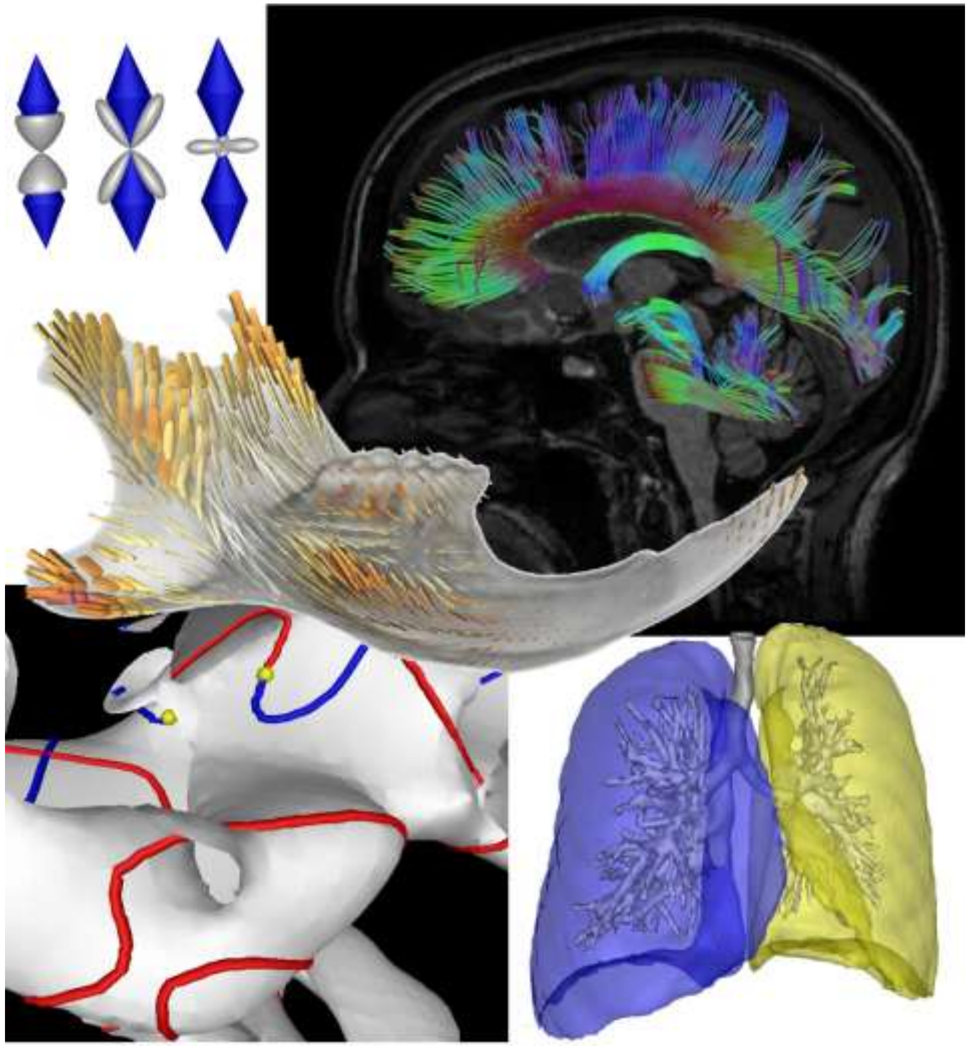
Bonn-Aachen International Center for Information Technology

1.1 Visualization: What and Why?

Visualization: What?

Visualization makes images out of data

- Some examples from our own work:



Visualization: Why?

“The ability to take data - to be able to understand it, to process it, to extract value from it, to visualize it, to communicate it - that’s going to be a hugely important skill in the next decades.”

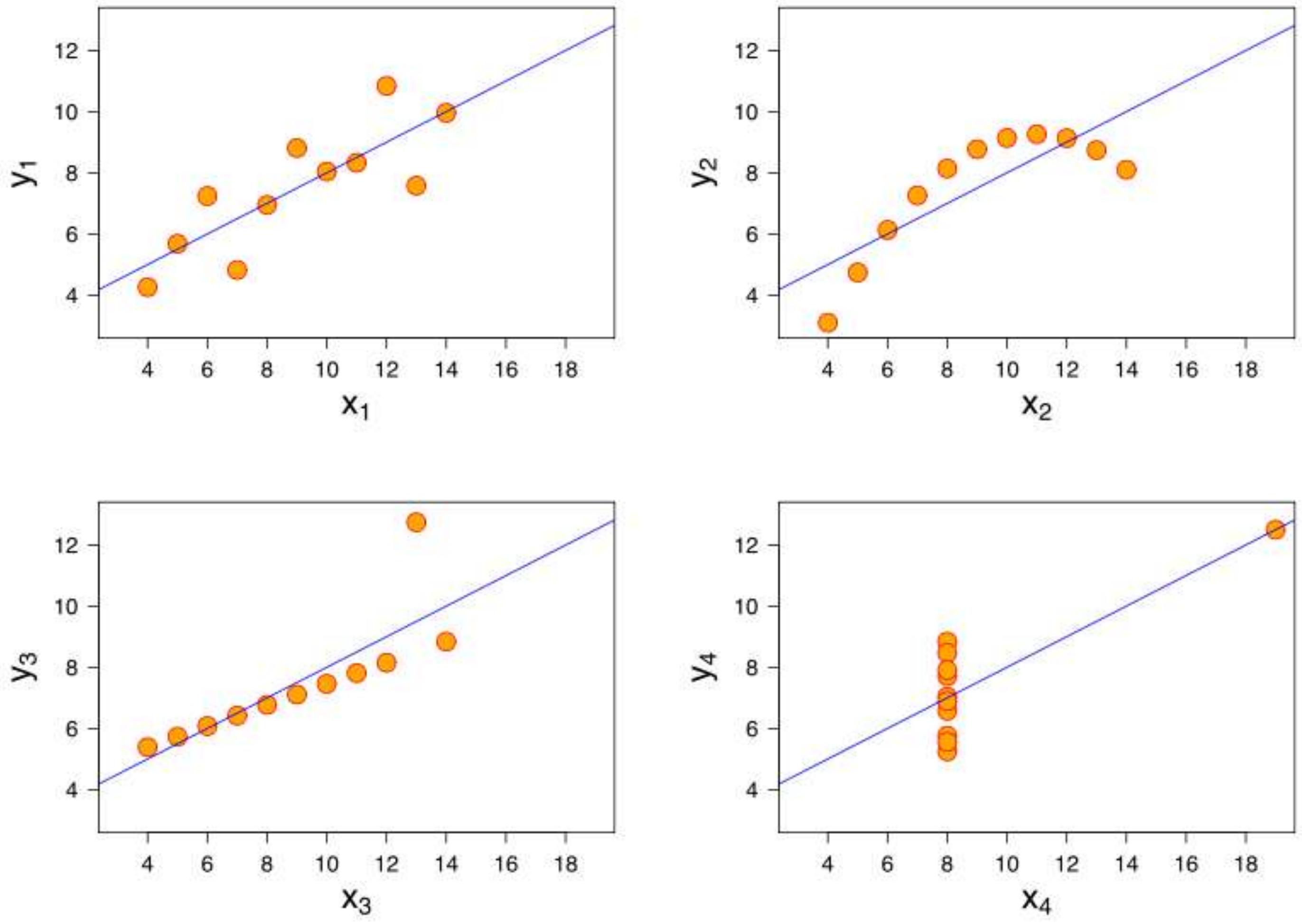
Hal R. Varian

John W. Tukey

“The best thing about bein **in visualization** s that you get to play in everyone's backyard.”

Visualization vs. Statistics

In 1973, Francis Anscombe constructed the "Anscombe Quartet", illustrating how different data with the same mean, variance, and linear regression can turn out to be when it is visualized



Anscombe’s Quartet Revisited

Animation from Matejka/Fitzmaurice (2017) based on Alberto Cairo’s “Datasaurus dozen”

Defining Visualization

- Definition by Oxford English Dictionary:
  - **to visualize:** to form a mental vision, image, or picture (of something not visible or present to sight, or of an abstraction); to make visible to the mind or imagination.
- In the words of Robert Spence (2007):
  - Visualization is solely a human cognitive activity and has nothing to do with computers

    Source: R. Spence, **Information Visualization,** Prentice Hall 2007
- To be more precise:
  - This lecture deals with **computer-supported data visualization**

Defining Visualization

- Definition by McCormick / DeFanti / Brown:
  - Visualization is a method of computing. It **transforms the symbolic into the geometric,** enabling researchers to observe their simulations and computations. Visualization offers a method for **seeing the unseen.** It enriches the process of scientific discovery and fosters profound and unexpected insights. In many fields it is already revolutionizing the way scientists do science.

Source: McCormick, B.H., T.A. DeFanti, M.D. Brown, **Visualization in Scientific Computing,** Computer Graphics Vol. **21**.6, November 1987

Defining Visualization

- Definition by Owen:
  - Visualization is essentially a **mapping process** from computer representations to perceptual representations, choosing encoding techniques to maximize human understanding and communication.

    Source: G.S. Owen, **HyperVis - Teaching Scientific Visualization Using Hypermedia.** ACM SIGGRAPH Education Committee 1999

Defining Visualization

- Definition by Haber / McNabb:
  - The use of computer imaging technology as a tool for **comprehending data** obtained by simulation or physical measurement by **integration of older technologies,** including computer graphics, image processing, computer vision, computer-aided design, geometric modeling, approximation theory, perceptual psychology, and user interface studies.

Footnote: R.B. Haber and D. A. McNabb, **Visualization Idioms: A Conceptual Model for Scientific Visualization Systems,** in Visualization in Scientific Computing, IEEE Computer Society Press 1990.

Visualization in Science

- Motivation by Friedhoff and Kiley:
  - The standard argument to promote scientific visualization is that today's researchers must consume ever higher volumes of numbers that gush, as if from a fire hose, out of supercomputer simulations or high-powered scientific instruments. If researchers try to read the data, usually presented as vast numeric matrices, they will take in the information at snail's pace. If the information is rendered graphically, however, **they can assimilate it at a much faster rate.**

Footnote: R.M. Friedhoff and T. Kiely, **The Eye of the Beholder,** Computer Graphics World, Vol. 13.8, p. 46, August 1990

Defining and Motivating Visualization

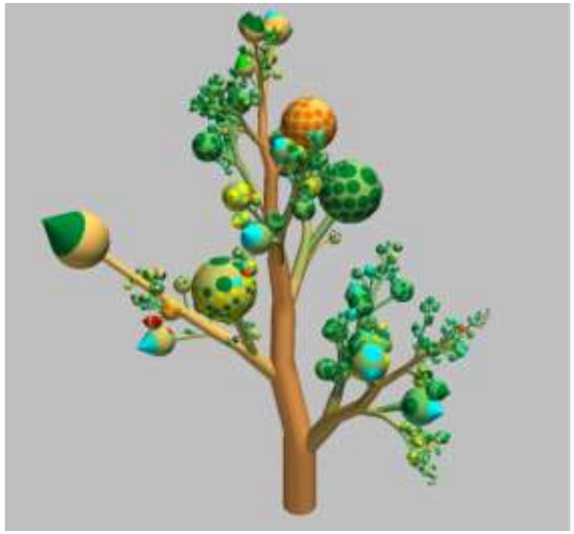
- Definition and motivation by Tamara Munzner:
  - **Computer-based visualization systems** provide visual representations of datasets designed to help people carry out tasks more effectively.
    Visualization is suitable when there is a need to **augment human capabilities** rather than replace people with computational decision-making methods.

Footnote: T. Munzner, **Visualization Analysis and Design,** A K Peters, 2015

- Quote attributed to Eugene Wigner (1902-1955):
  - It is nice to know that the computer understands the problem.
    But I would like to understand it too.

The Value of Visualization

- [van Wijk 2005] quantifies the profit from a visualization tool as a **technology** with which n users each analyze m datasets as
$$F = nm(W(\Delta K) - C_s - kC_e) - C_i - nC_u$$
  – Accounts for the value W of knowledge gain ΔK, but also the cost C_s of preparing the data, C_e of exploring it over k rounds, C_u of learning to use the tool, and C_i of building it
- Visualization can also be seen as an **art** and a **science**

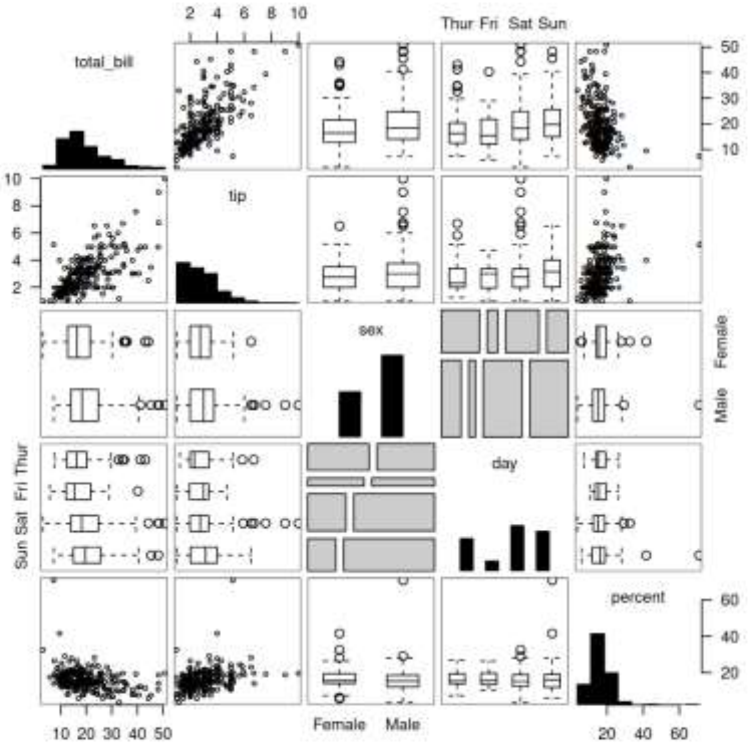


Goals of Visualization

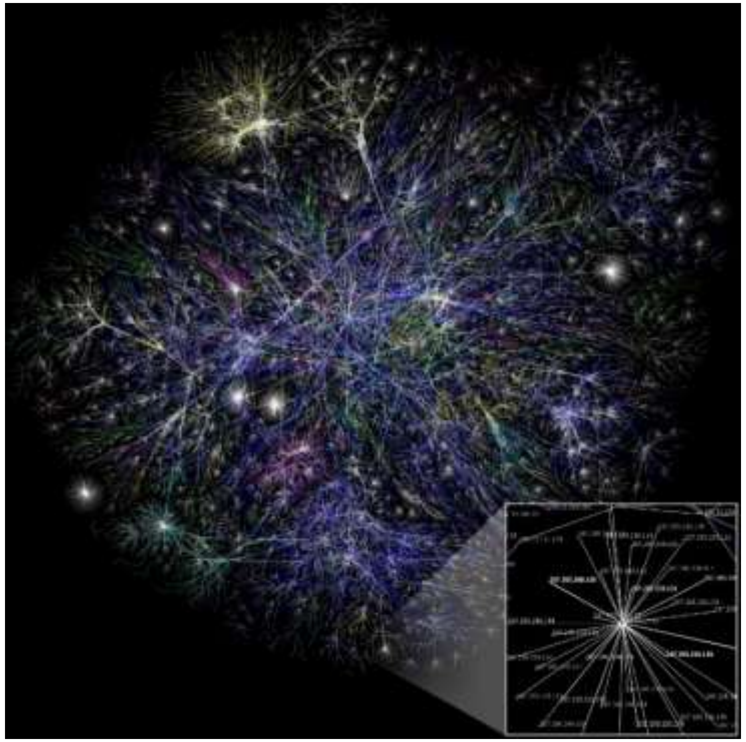
- **Analysis** leading to insight
  - Extract information from data
  - Discover new knowledge, generate hypotheses
- **Communication** and education
  - Present findings to others and discuss them
  - Teach non-experts
- **Steering**
  - Interactively control and drive simulations or longer-term measurements
  - Fast data analysis, get to insight sooner

Information Visualization

Focuses on abstract (non-spatial, not necessarily numerical, fundamentally discrete) information.



Tabular Data



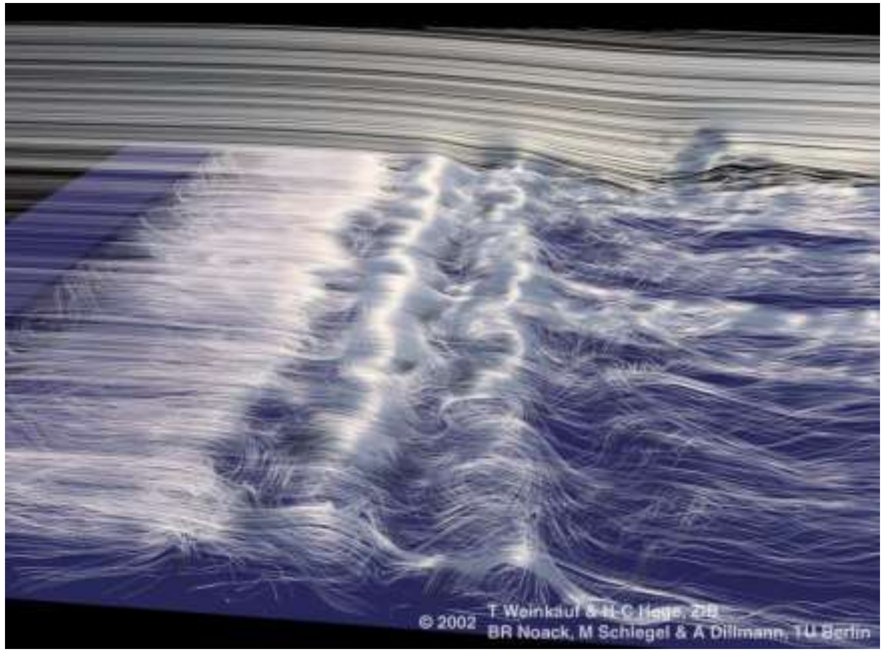
Graphs

Maps

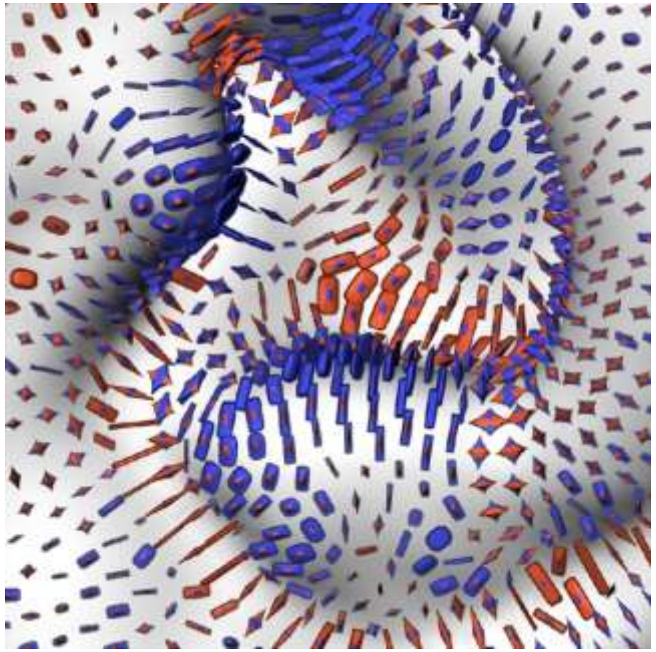
Scientific Visualization

Focuses on scientific and medical data (2D/3D space +time, +parameter) from measurements or simulations.

Scalar Fields



Vector Fields

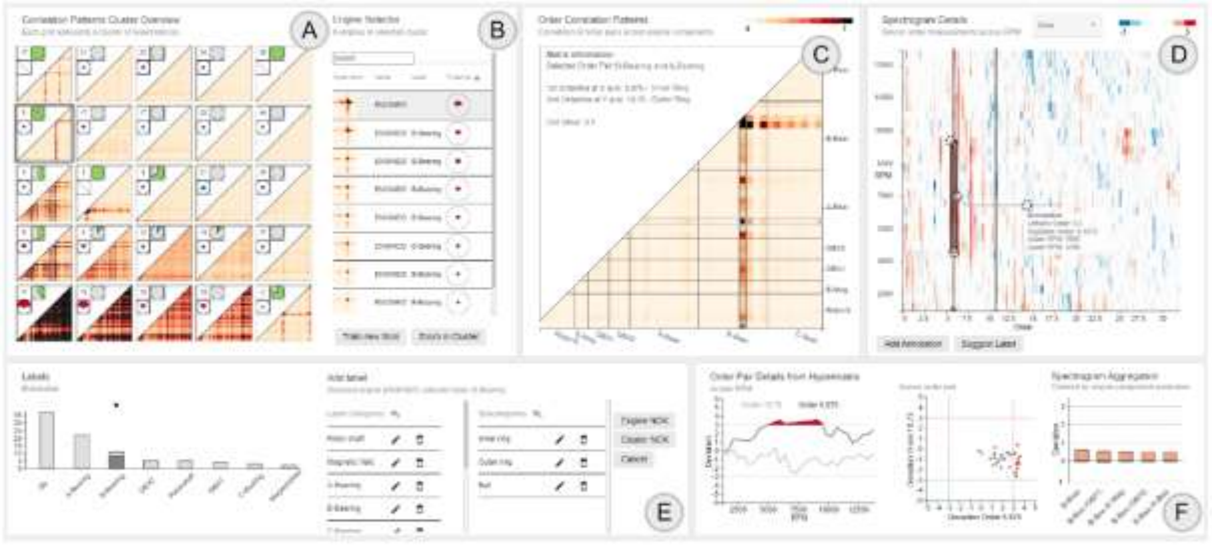


Tensor Fields

Visual Analytics

Focuses on integrating automated analysis (e.g., data mining / machine learning) with interactive visualization to facilitate analytical reasoning about large and complex data.

Decision making based on EHRs
[Cheng et al., VIS 2021]



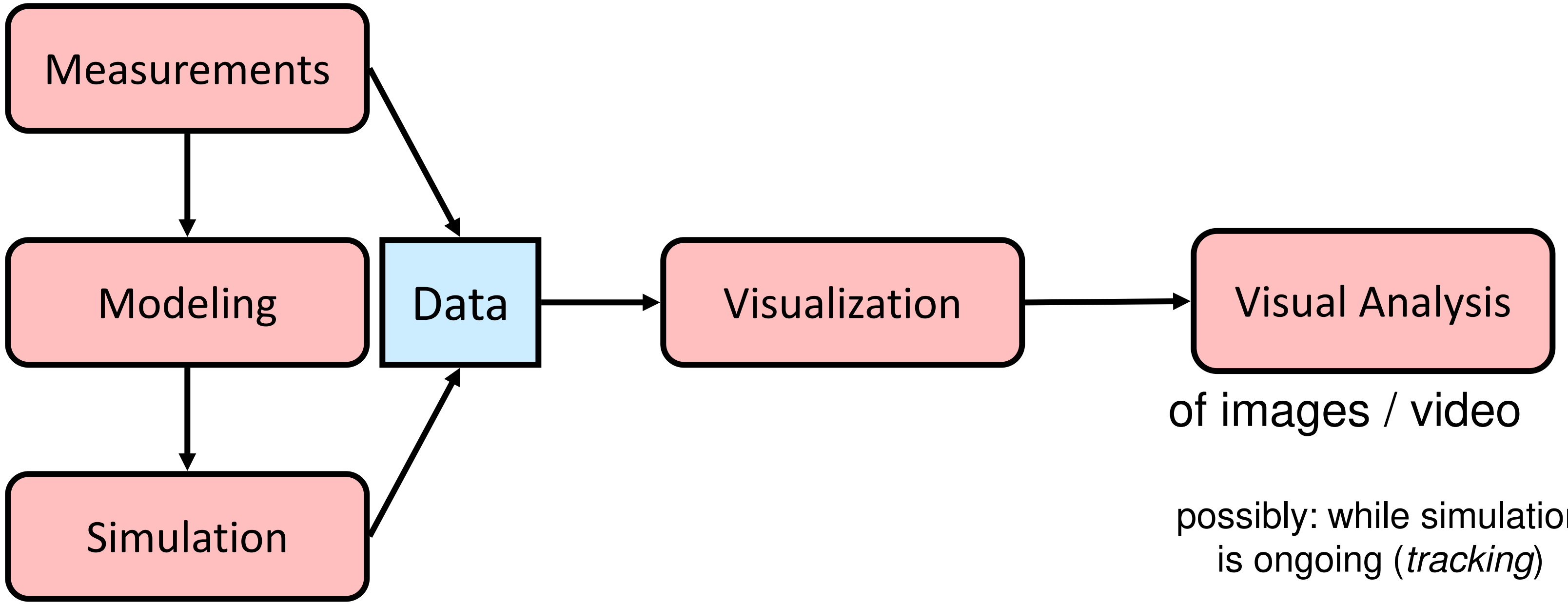
Quality control in serial manufacture
[Eirich et al., VIS 2021]

Explaining Visual Analytics

- Explanation by Daniel Keim:
  - **Visual Analytics** methods allow decision makers to combine their human flexibility, creativity, and background knowledge with the enormous storage and processing capacities of today’s computers to gain insight into complex problems. Using advanced visual interfaces, humans may directly interact with the data analysis capabilities of today’s computer, allowing them to make well-informed decisions in complex situations.

Source: visual-analytics.eu – the new visual analytics portal, 2017

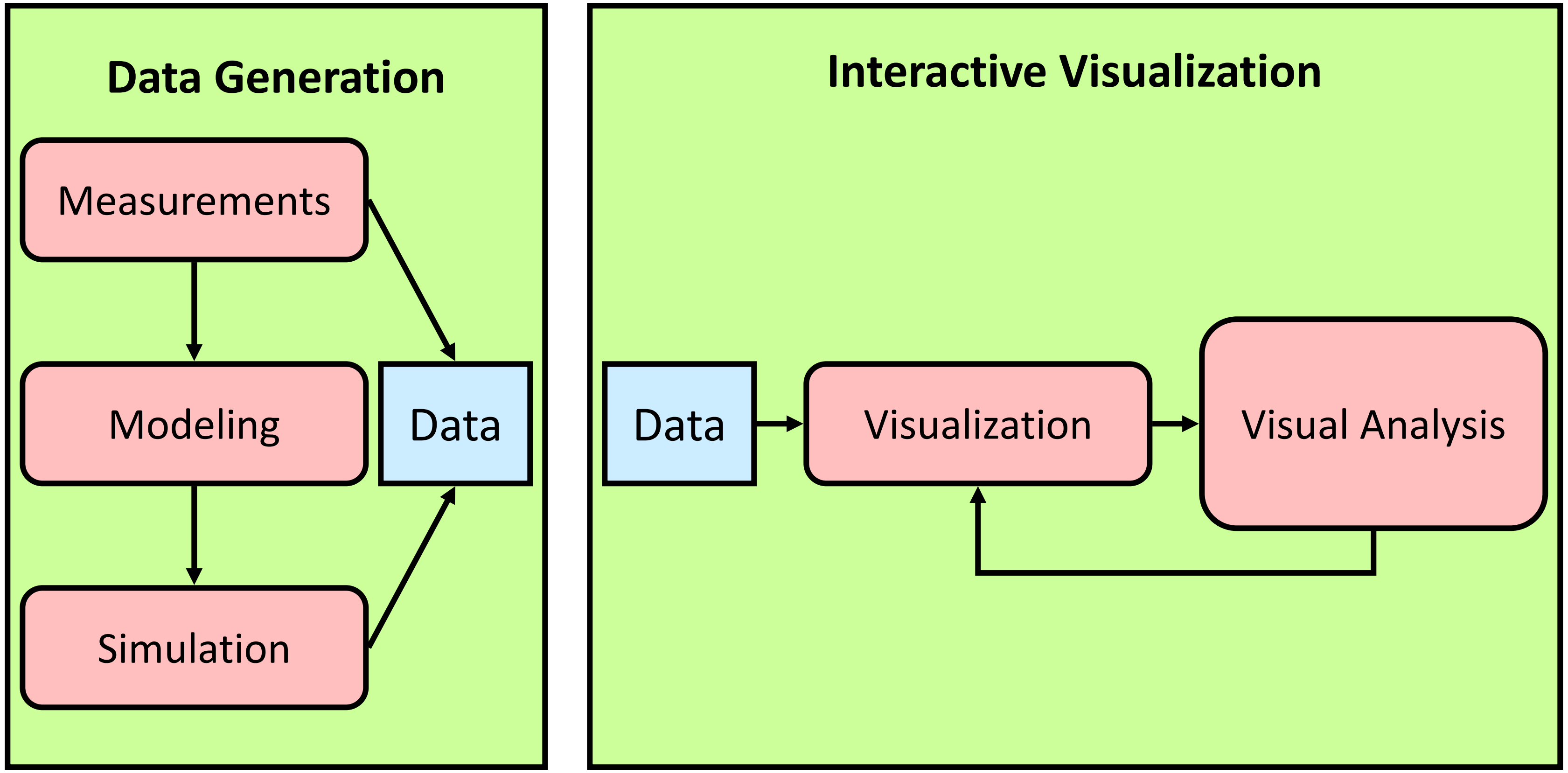
Non-Interactive Visualization



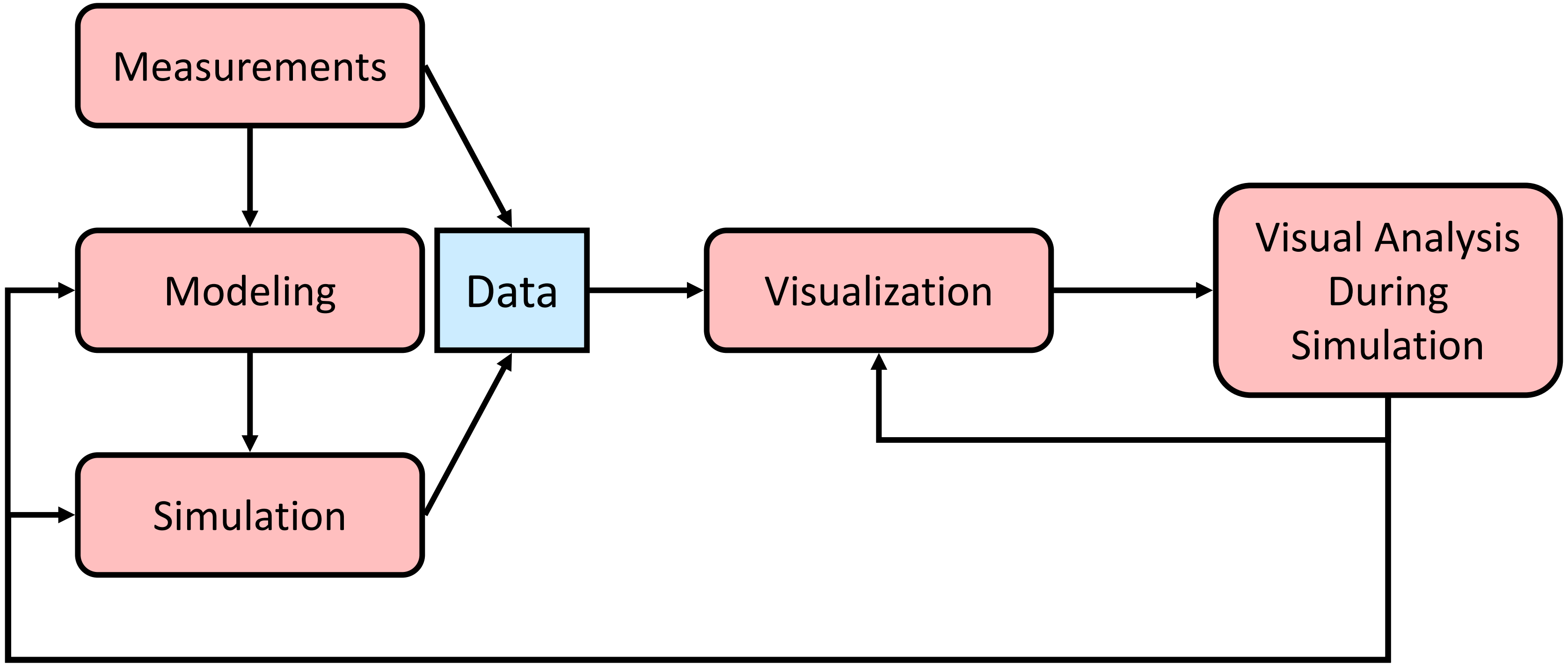
of images / video

possibly: while simulation is ongoing (*tracking*)

Interactive Visualization as Post-Process

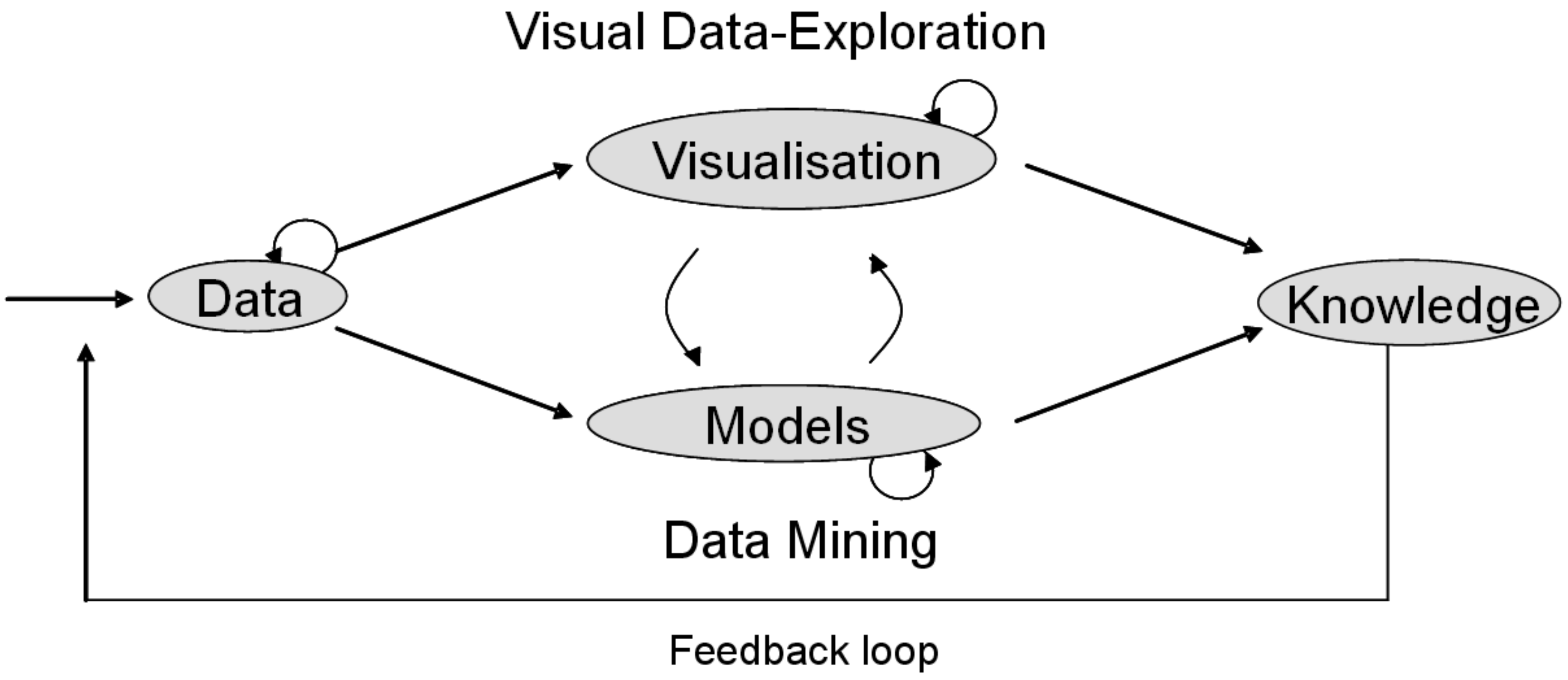


Computational Steering



Visual Analytics

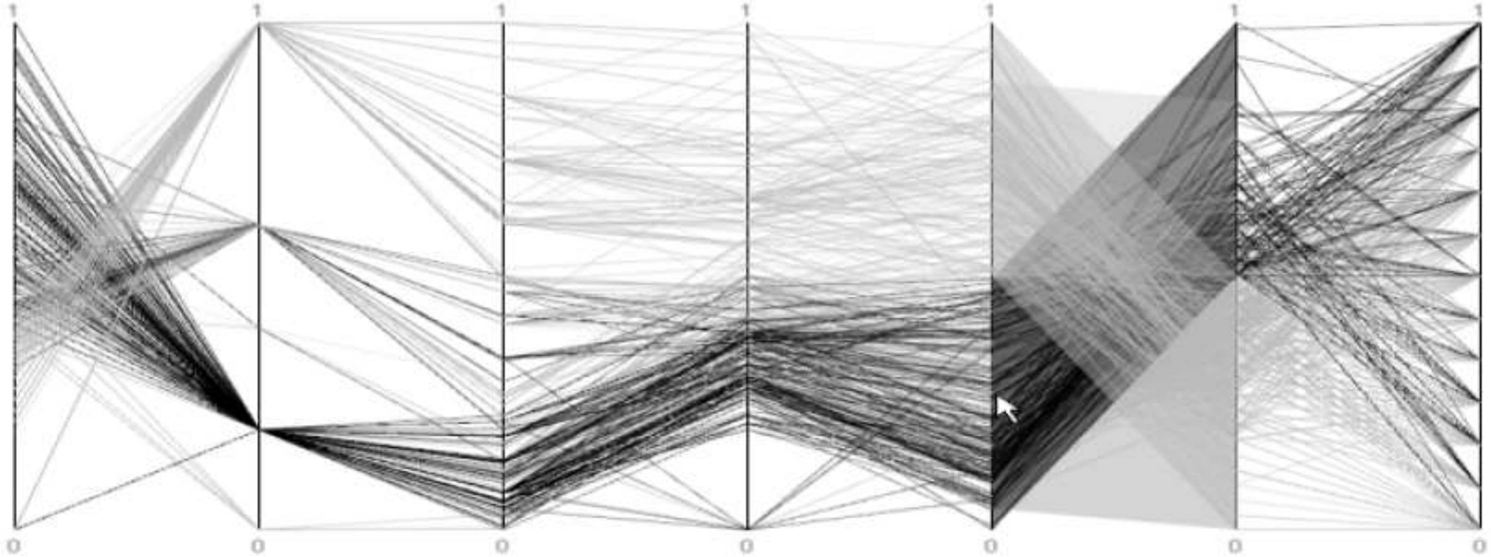
- Source: Keim et al. 2008

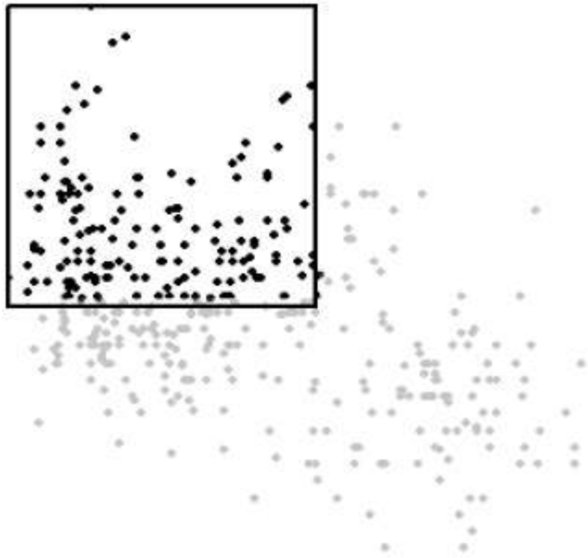


1.2 A Preview of This Lecture

Visualizing Multidimensional Data

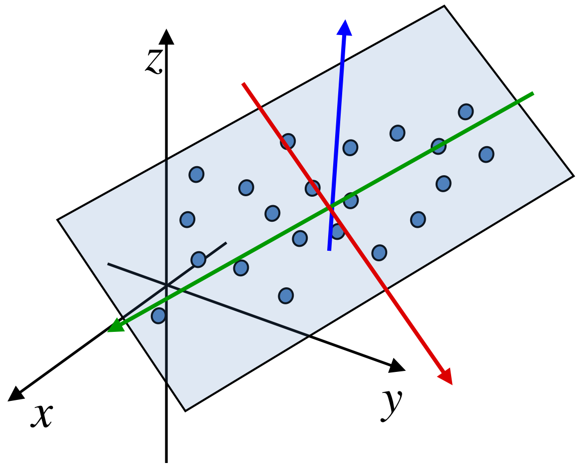
- Going from standard visualizations to visualizing data that has **more than three dimensions**
  - Visual encodings
  - Interaction techniques
  - User guidance

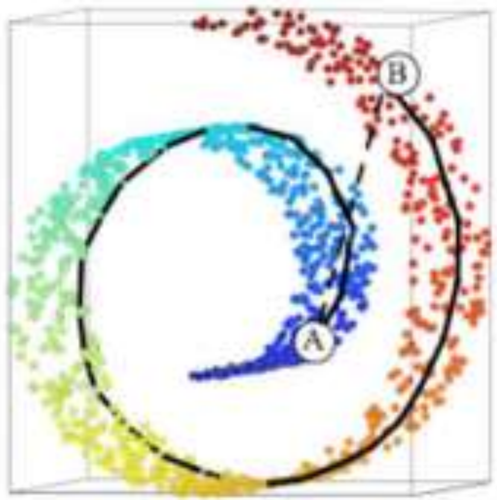


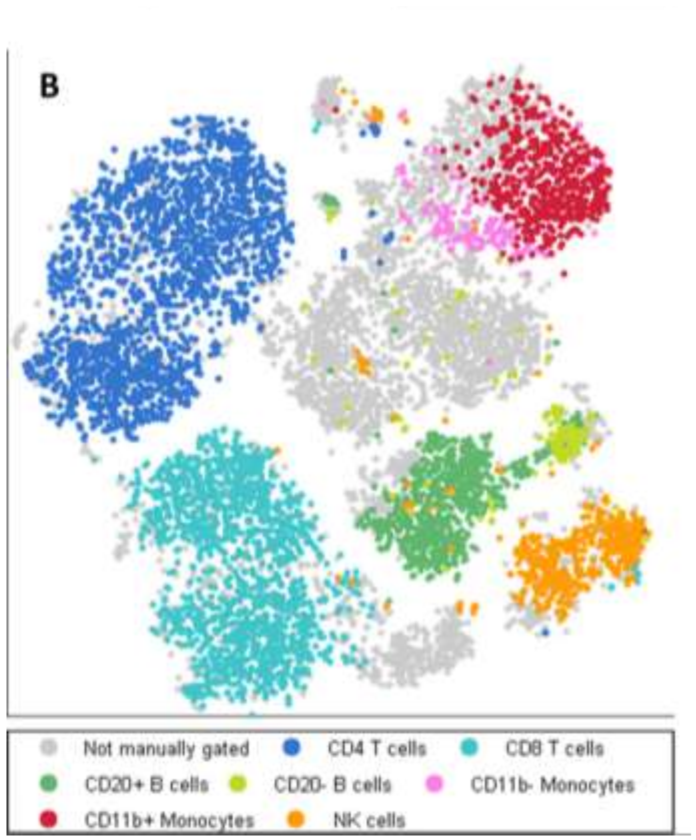


Dimensionality Reduction

- **Embed high-dimensional data into a lower-dimensional space** in a way that preserves important structure
  - From linear techniques to manifold learning
  - Unsupervised and supervised embeddings
  - Distance vs. cluster preserving embeddings

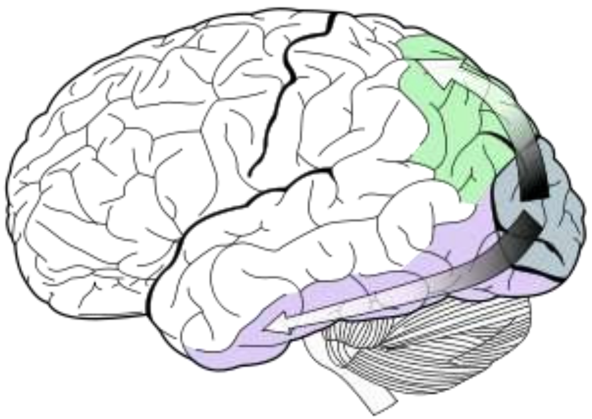


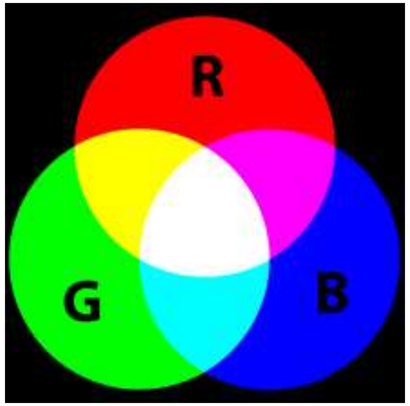


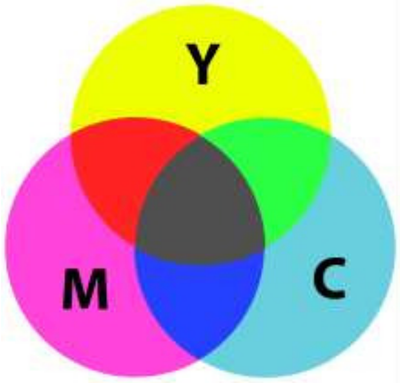


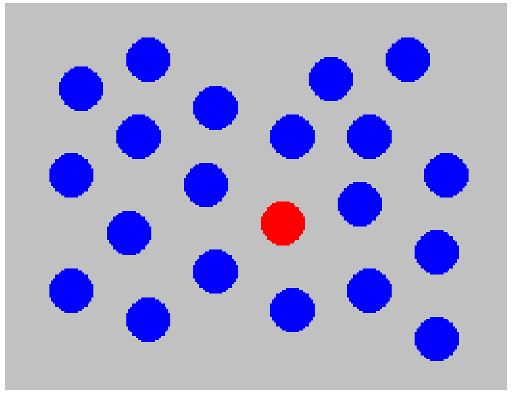
Color and Perception

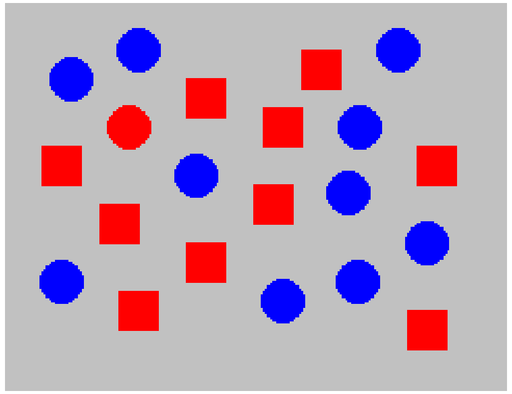
- Basics of **human (visual) perception**
- Different ways of **representing color**
- Implications for **data visualization**







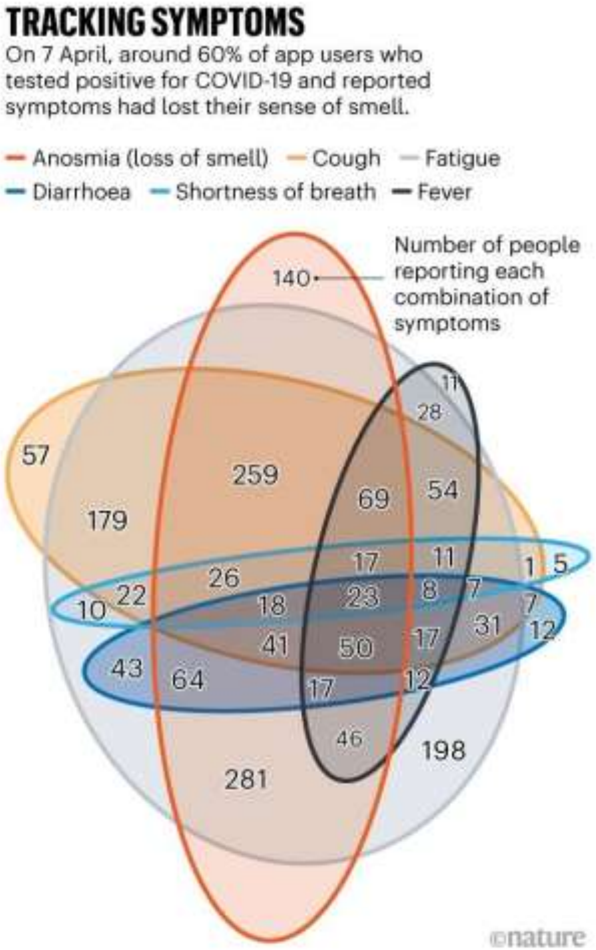




Visualization Design

- Rules (of thumb) on how to **design and develop effective visualizations** and **visual analytics systems**

Image from: Nature News, April 22, 2020



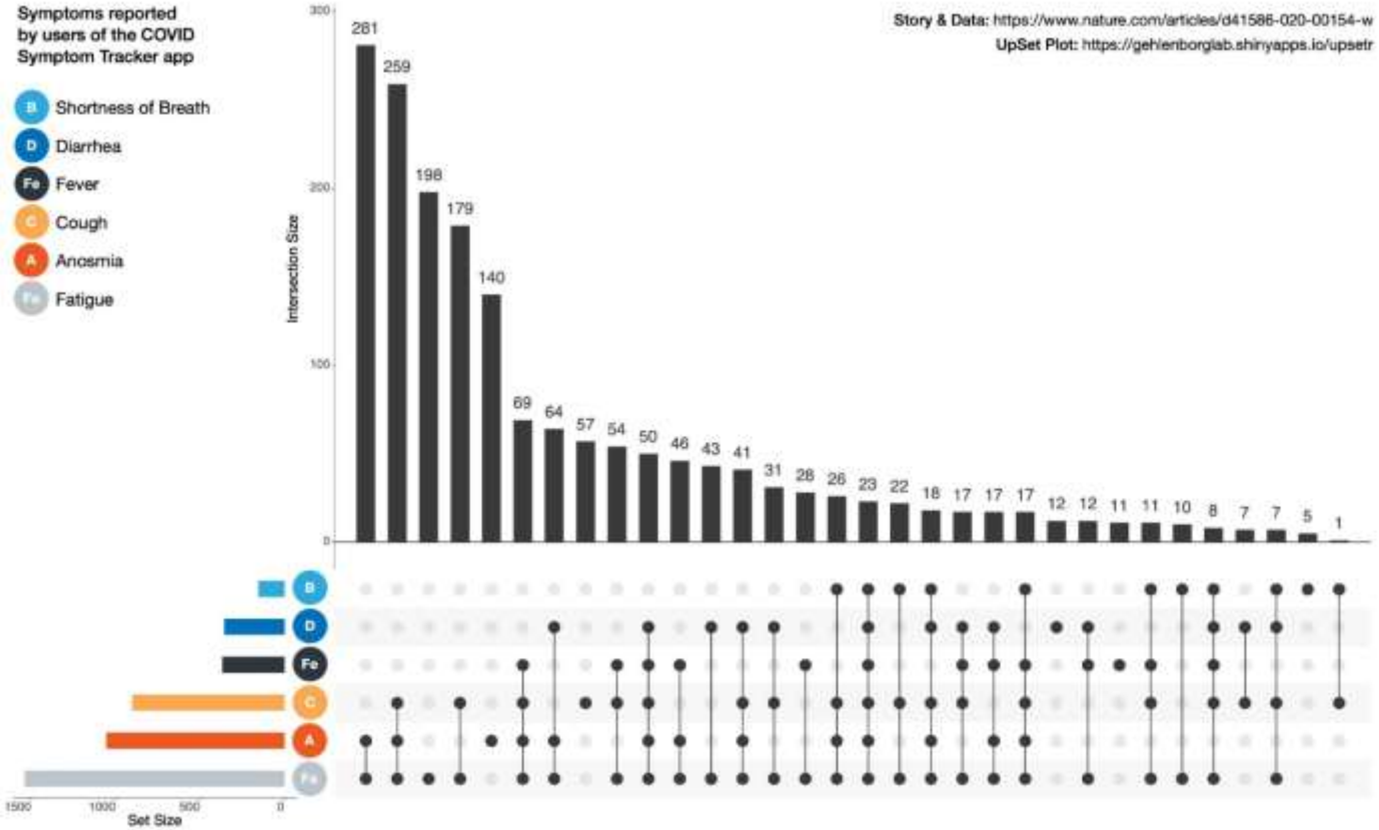
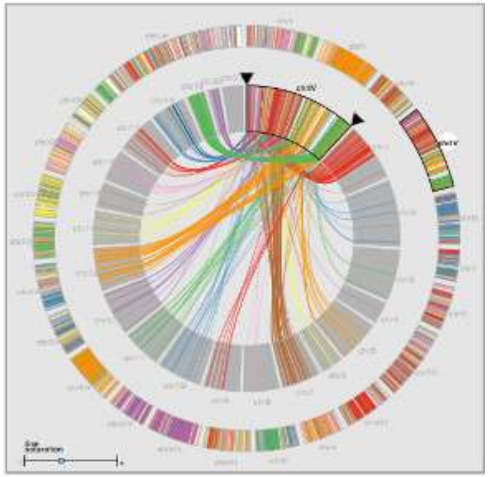
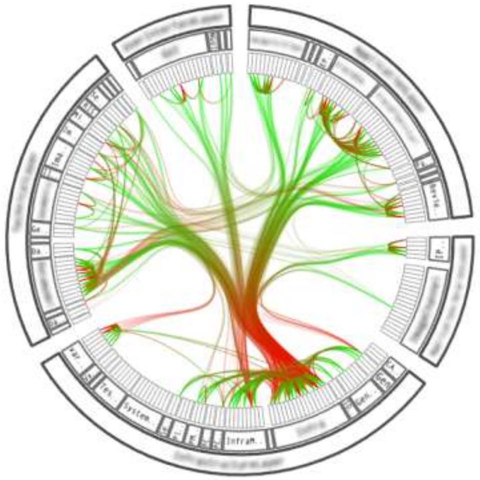


Image from Nils Gehlenborg, Harvard

Graph Visualization

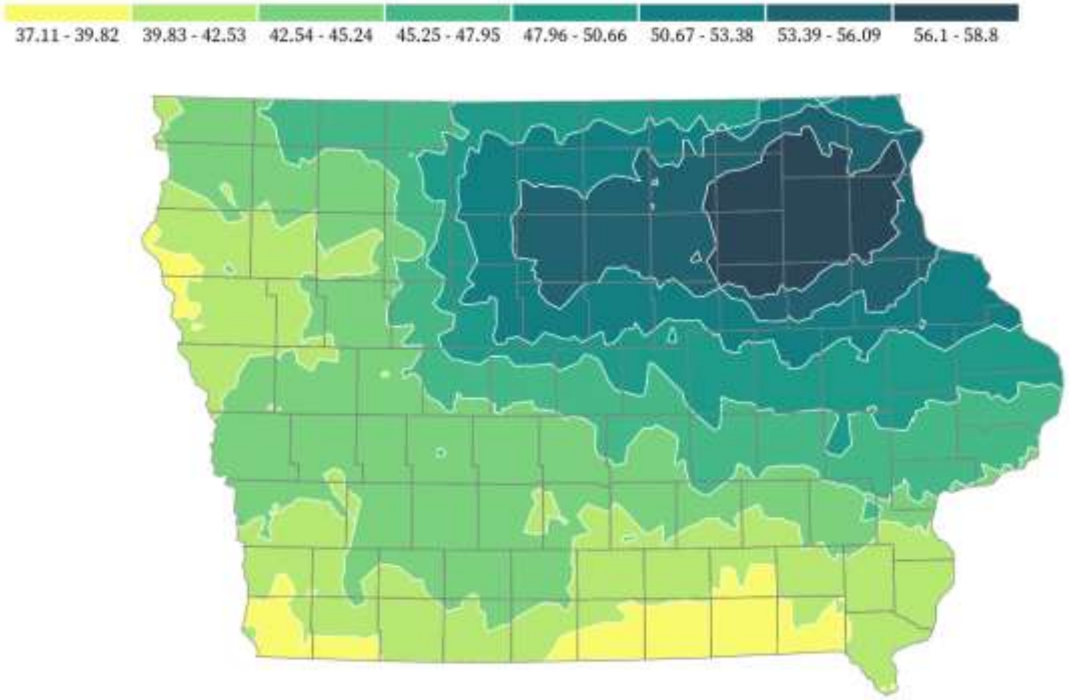
- Visualize **trees and general graph structures**





Geospatial Data Visualization

- Different **map projections** and their properties
- Point, line, and area based **visual encodings**
  - Scattered data interpolation

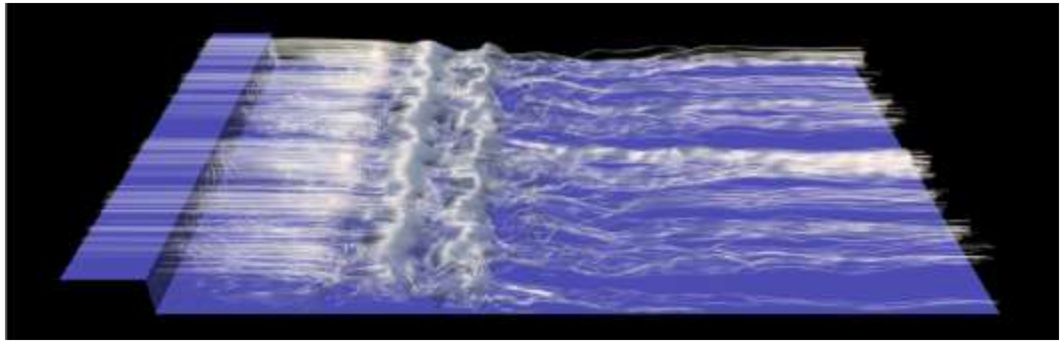


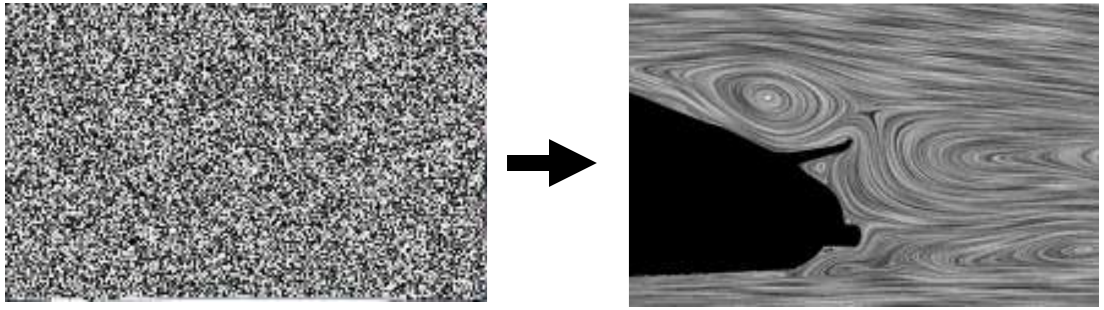
Volume Visualization

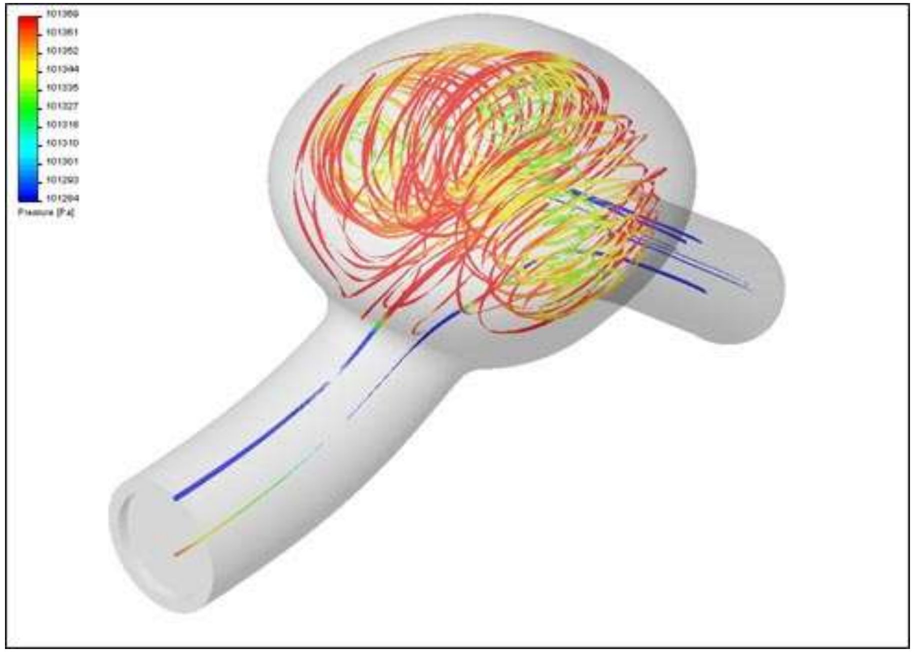
- Produce images from **three-dimensional data volumes**
  - Generate surfaces that represent objects
  - Depict gases, fluids
  - Semi-transparent renderings of nested objects

Vector Field Visualization

- Depict the **flow of fluids or air**
  - Trace the trajectories of fluid particles
  - Turn random noise into meaningful pictures
  - Applications in industrial design, climate research, medicine etc.

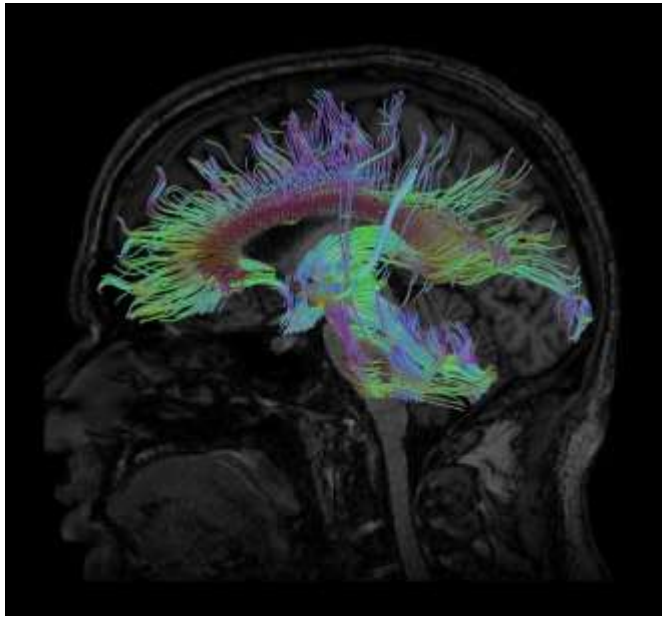


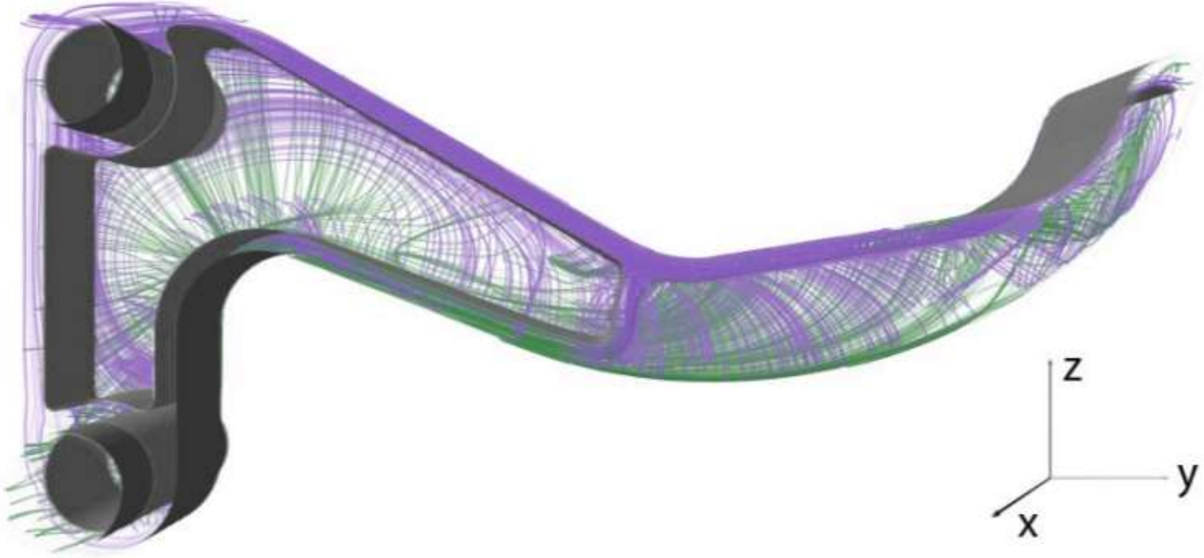




Tensor Field Visualization

- Visualize **matrix fields**
  - Applications include
    - brain imaging (heat motion of water)
    - engineering (elastic properties of materials)
  - Develop a geometric intuition for linear algebra

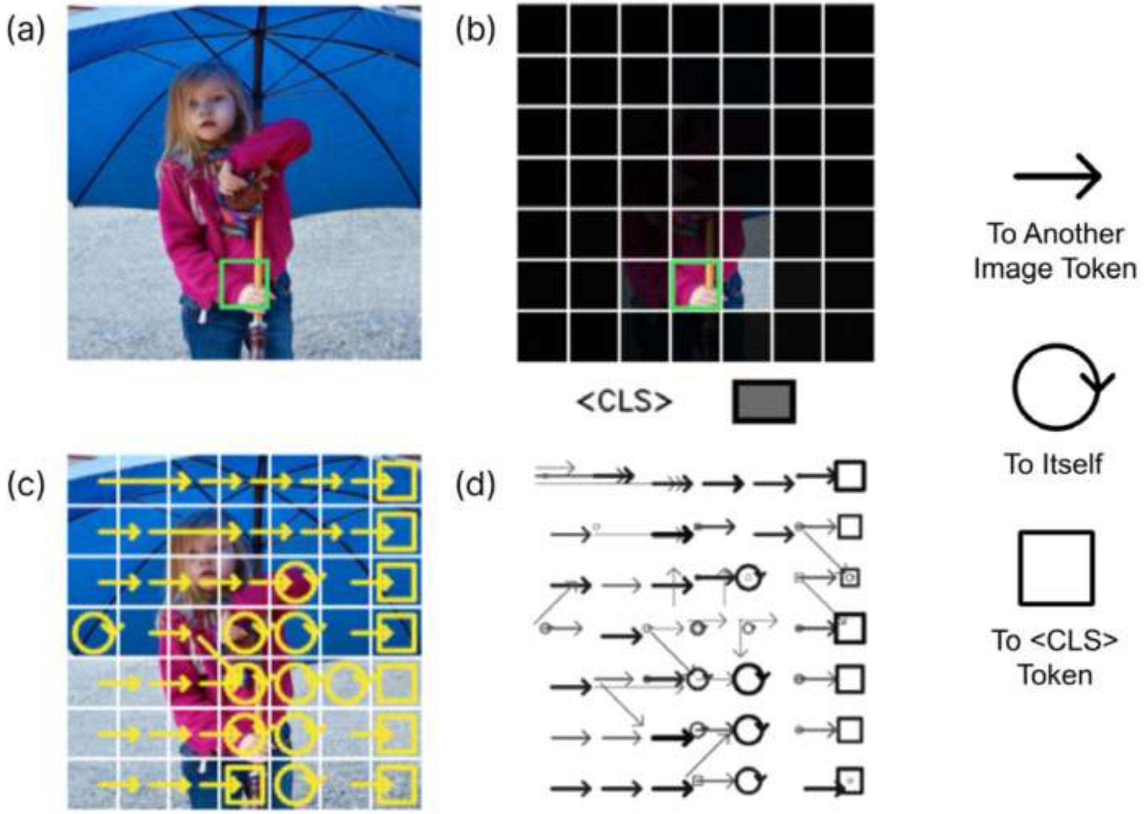




Visualizing Neural Networks

- Use visualization to better understand what happens within **deep neural networks**
  - *Special focus:* Image classification

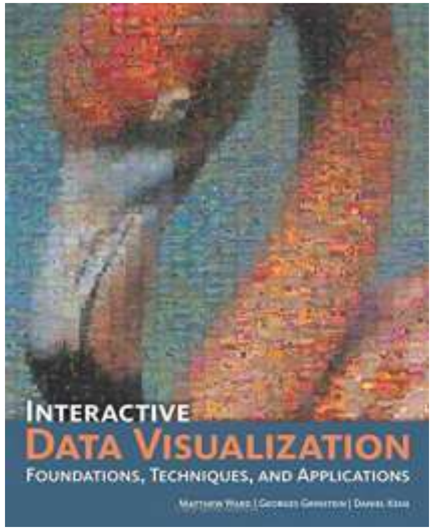
[Selvaraju et al. 2019]



[Yeh et al., TVCG 2024]

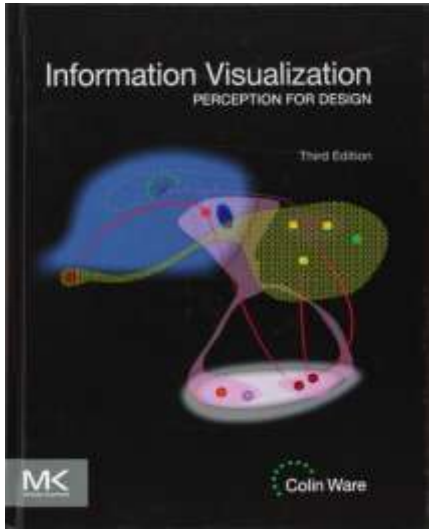
Books for First Part of Lecture

All required information is given during the lecture, and much is taken from research papers. Optionally, consulting the following textbooks will broaden and deepen your knowledge:



- M. Ward et al., **Interactive Data Visualization: Foundations, Techniques, and Applications.** CRC Press, 2010

For perceptual aspects:

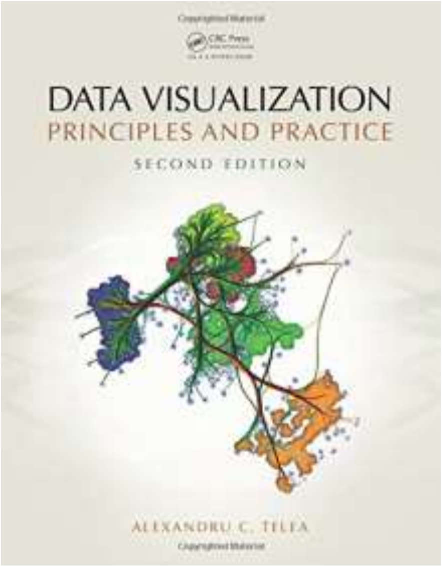


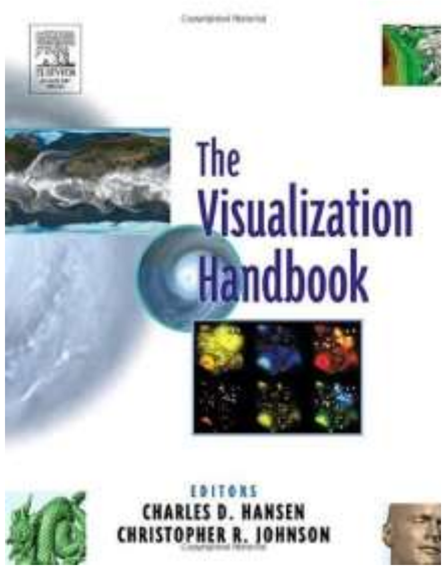
- C. Ware, **Information Visualization. Perception for Design**, Morgan Kaufmann, 3rd edition, 2013

More Recommended Books

Books relevant for large parts of the lecture, available via the university library

- A.C. Telea, **Data Visualization: Principles and Practice**, CRC Press, 2nd edition, 2015
- C.D. Hansen, C. Johnson, **Visualization Handbook**, Academic Press, 2004

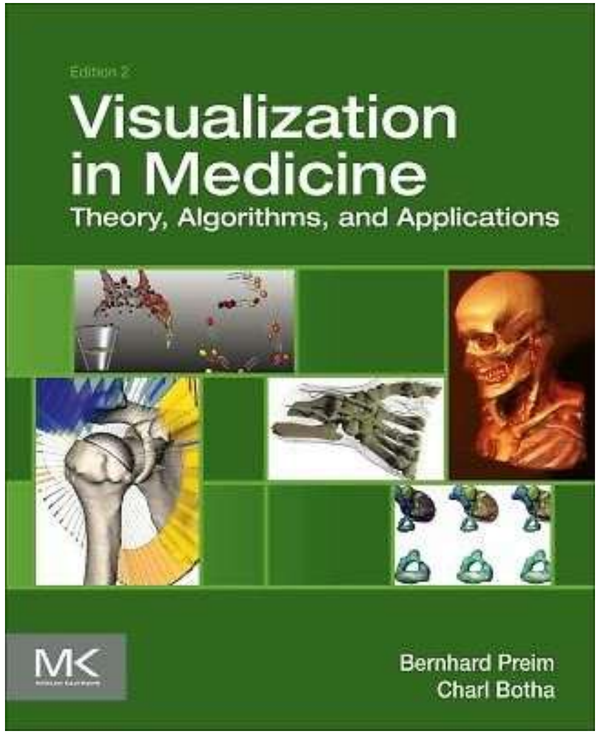




Recommended e-Book

E-Book relevant for several parts of the lecture:

- B. Preim, C. Botha, **Visual Computing for Medicine: Theory, Algorithms, and Applications**
  Morgan Kaufmann, 2014
  - Available online *from within university network*
  - Find link to electronic version via http://www.ulb.uni-bonn.de/



1.3 Lecture Organization

People

- **Lecture:**
  Prof. Dr. Thomas Schultz
  schultz@cs.uni-bonn.de

- **Exercises:**
  Annika Mikliss
  mikliss@cs.uni-bonn.de

Lecture Times

- **Lectures** will be held face-to-face
  - Tuesday 10:15-11:45 in **HS3 HSZ CP** (Friedrich-Hirzebruch-Allee **5**)
  - Wednesday 10:15-11:45 in **0.109 b-it** (b-it, Friedrich-Hirzebruch-Allee **6**)
  - ZOOM or video lectures will replace face-to-face if necessary
- Slides, assignments, additional communication via **eCampus**
  - Please sign up there as an attendee!
- I am planning to make **lecture recordings** available

Exercises

- **12 Exercise Sheets**
  - Includes “setup sheet” (only bonus points) and mock exam (no credit)
  - Balanced programming, experiments, theory, paper reading
  - Programming in Python + packages
  - Experiments in Paraview
  - You will have at least one week to work on each (regular) sheet
    - Assigned on Thursday, return by **Friday 10:00**
    - Please note that the “setup sheet” and the first regular sheet overlap!
- Exercise classes on Wed 14:15-15:45 in 0.109 b-it
  - General attendance is *not* mandatory, **but** you will have to present at least once during the semester

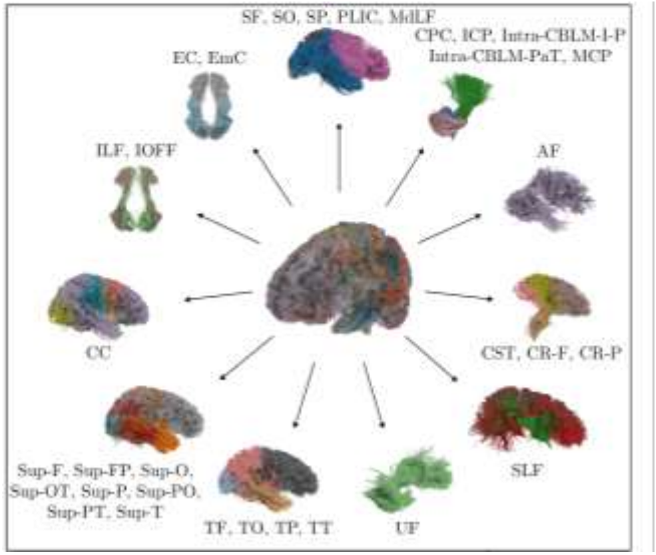
Rules for Exercises

- **Computer Science** students need 50% of the points from exercises (and present once) for admission to the exam
- **MI / HCIS** students only need 35% (and present once)
  - Compensates difference in ECTS
- **Bonus questions** allow you to get more than 100%
- Please work in **teams of three.**
- **Plagiarism is not acceptable.** We will not give credit for solutions that have obviously been copied.
- **Generative AI:** Can be used to help you further your own understanding, but copy-pasting exercises/solutions does not help you build skills and constitutes academic fraud.

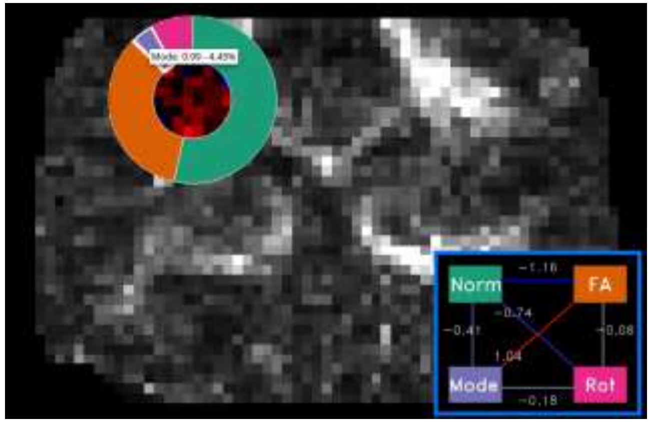
Credit

- *Tentative (unconfirmed)* dates for the **written exam** are:
  - August 3
  - September 21
- This course is cross-listed in two programs:
  - **Computer Science** at University of Bonn: 9 ECTS
  - **MI / HCIS** at the b-it: 8 ECTS
    - Different assumptions about workload for exercises
    - RWTH students require *fewer points from homeworks*
  - Do we have students from other programs?

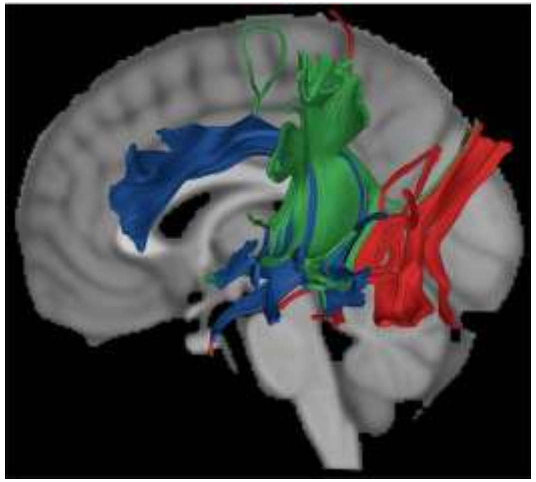
Consider Doing Your Thesis With Us!



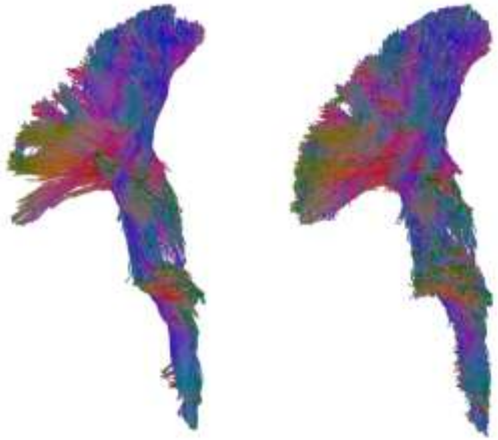
[Bisten et al., ImagNeuro 2026]



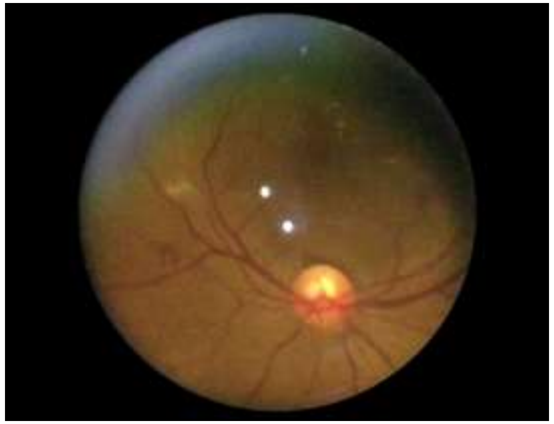
[Bareth et al., EuroVis SP 2023]



[Khatami et al., MedIA 2022]



[Grün et al., VCBM 2021]

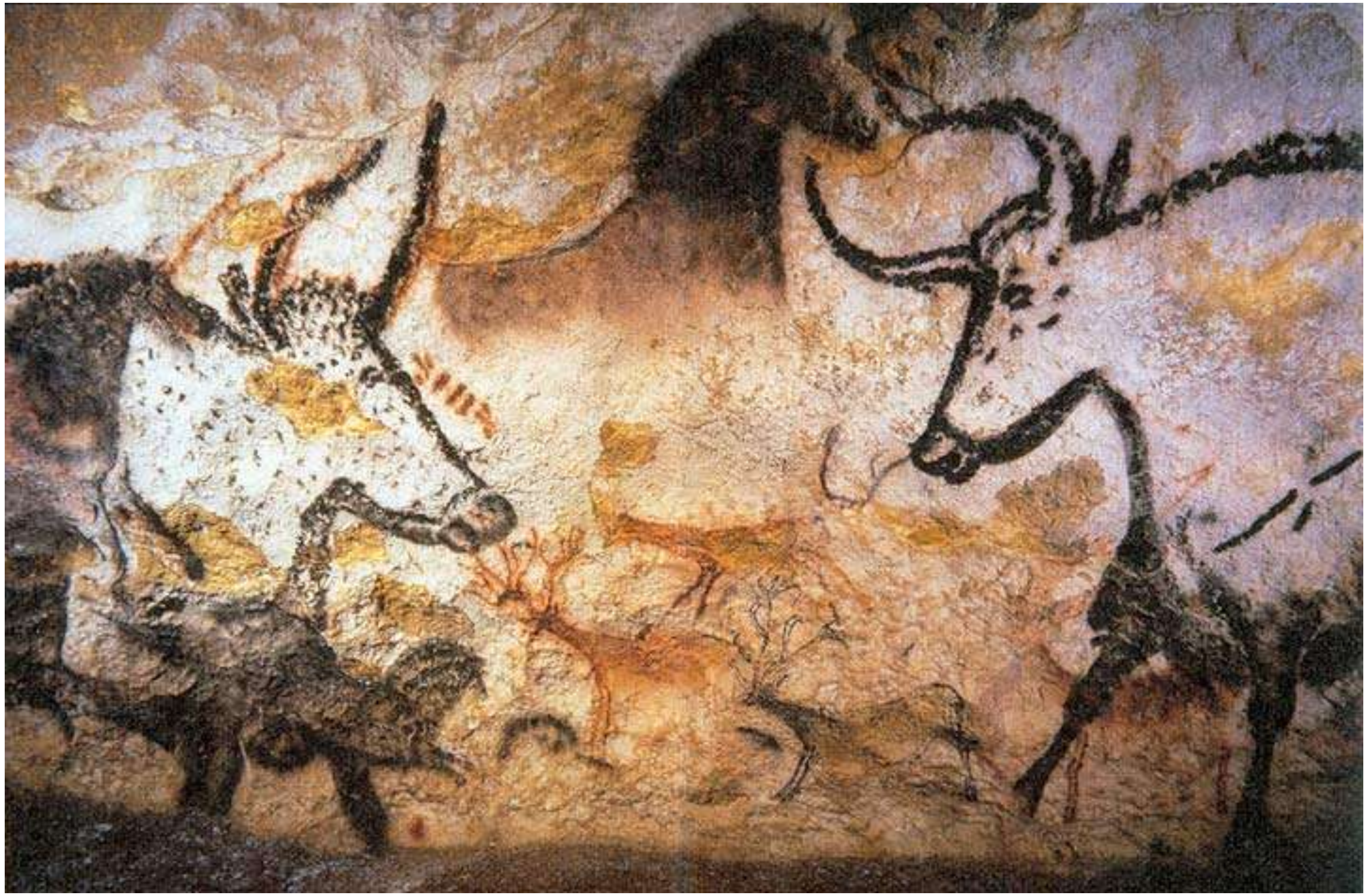


[Mueller et al., OMIA 2020]

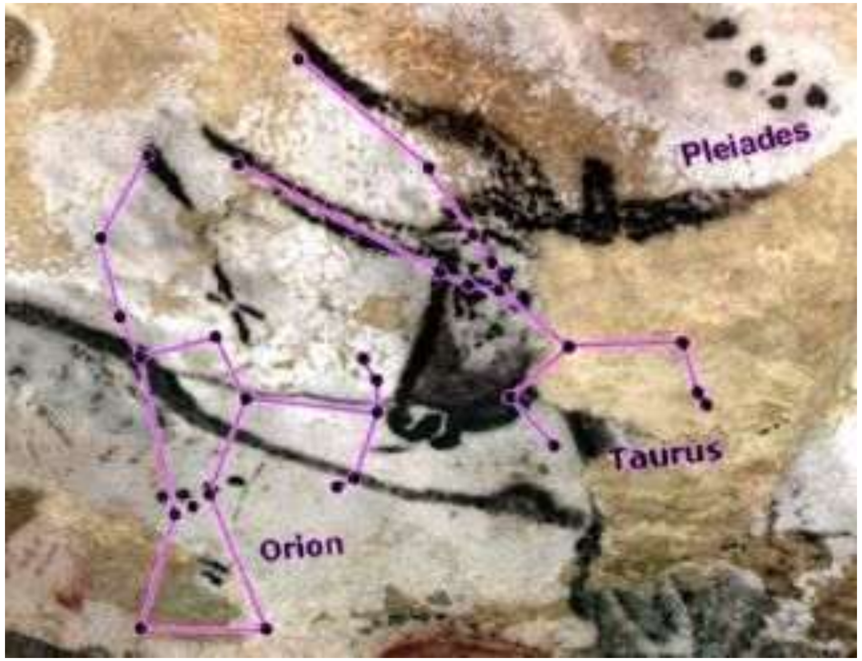
[Soundararajan et al., EuroVis 2015]

1.4 History of Visualization

Earliest Visualizations

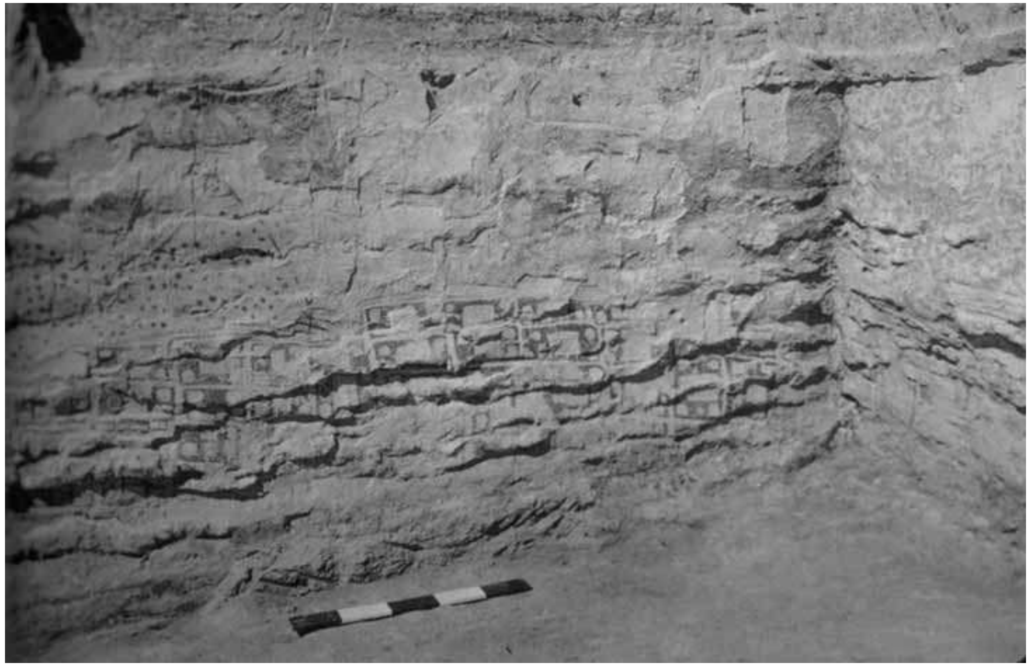


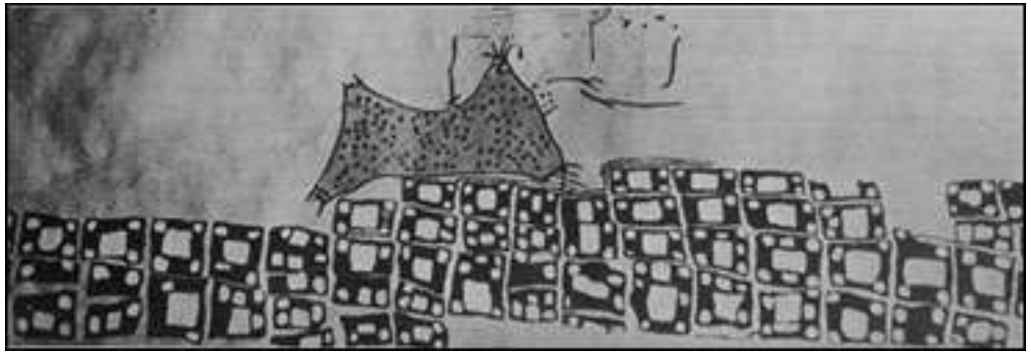
Lascaux Cave, around 17,000 years ago



A Paleolithic Star Chart?

Early Maps on Stone





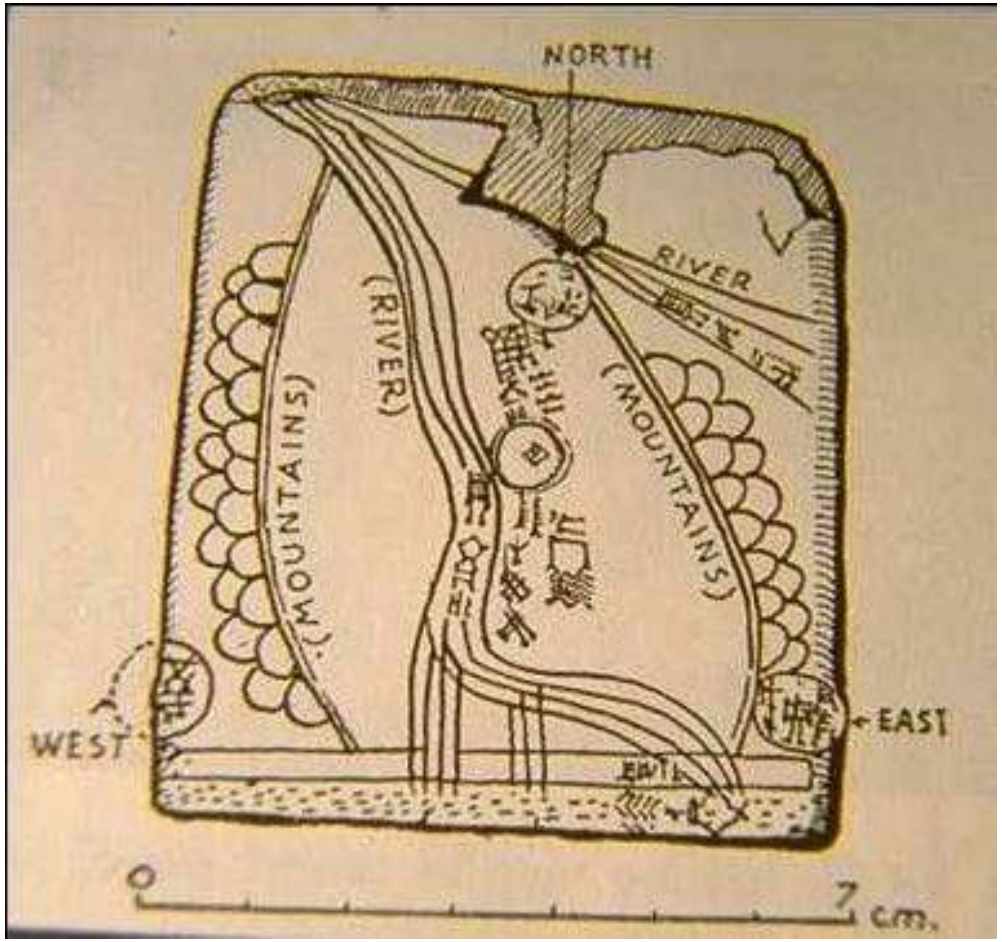
Map of Catal Höyük

6200 BC

Excavation

Reconstruction

Early Maps on Clay



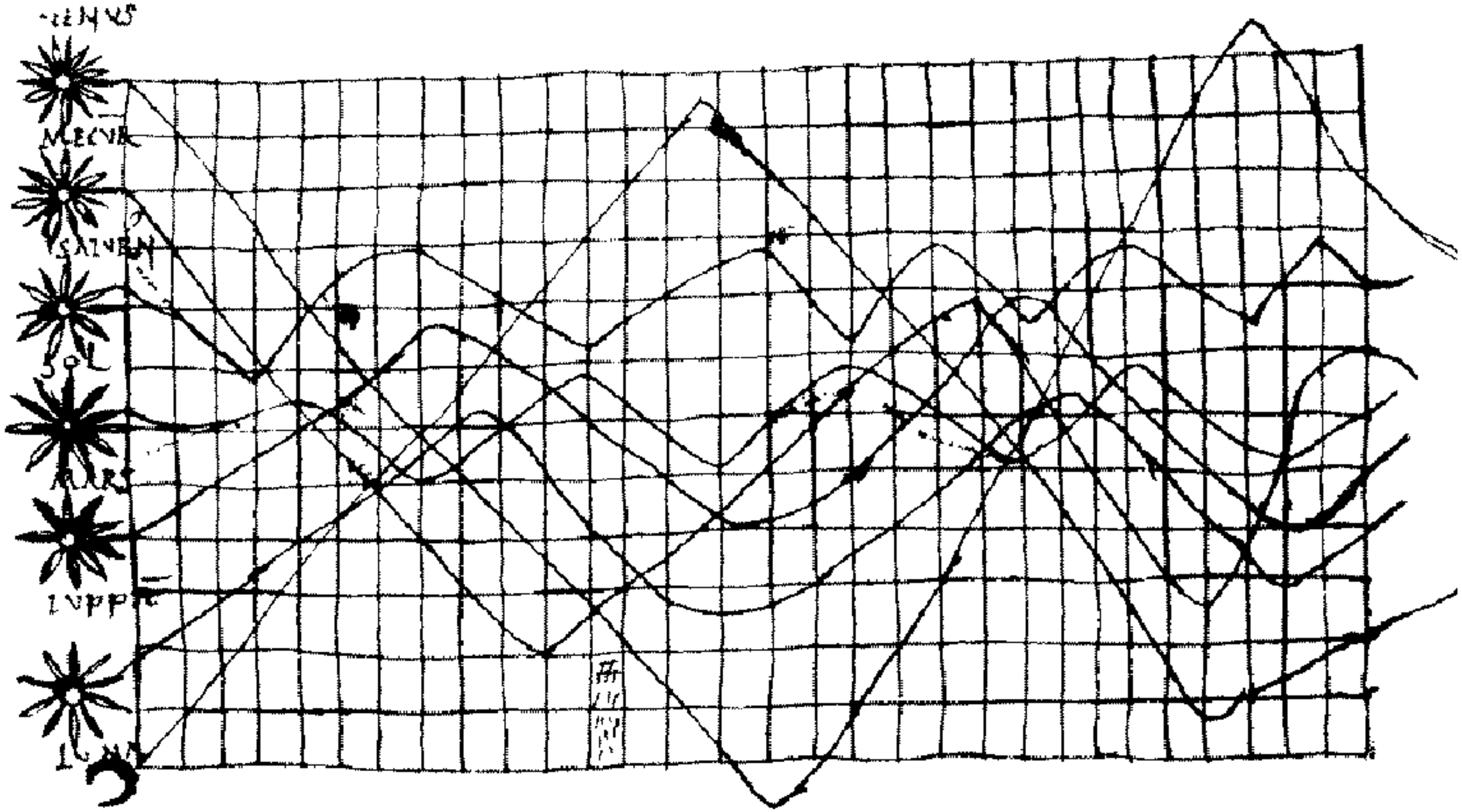
Map of Ga-Sur

2500 BC

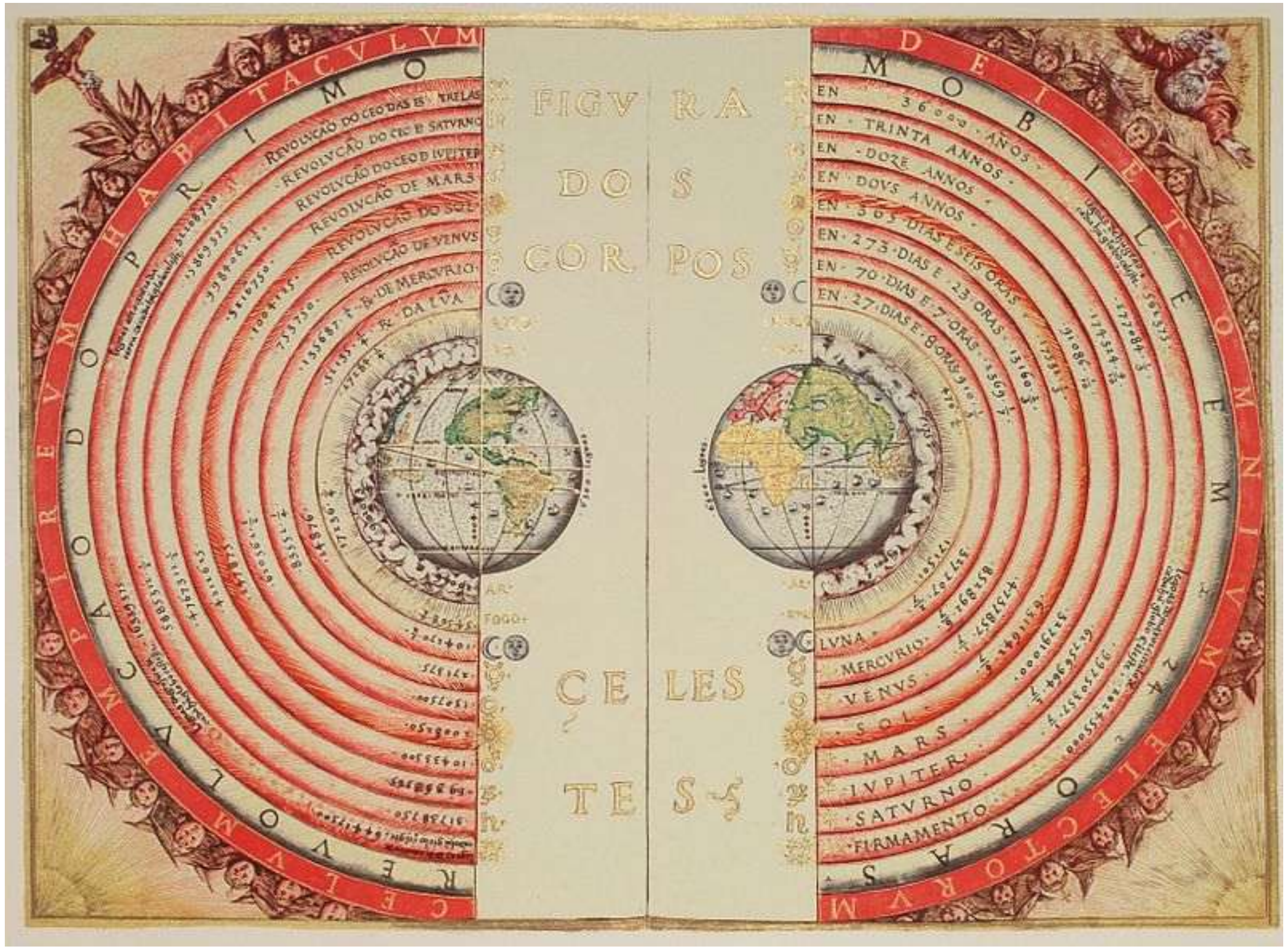
- Map on clay
- Reconstruction

Early Graphs of Time Series

- Time series: inclination of planets
  - Early use of a coordinate grid (10th century)



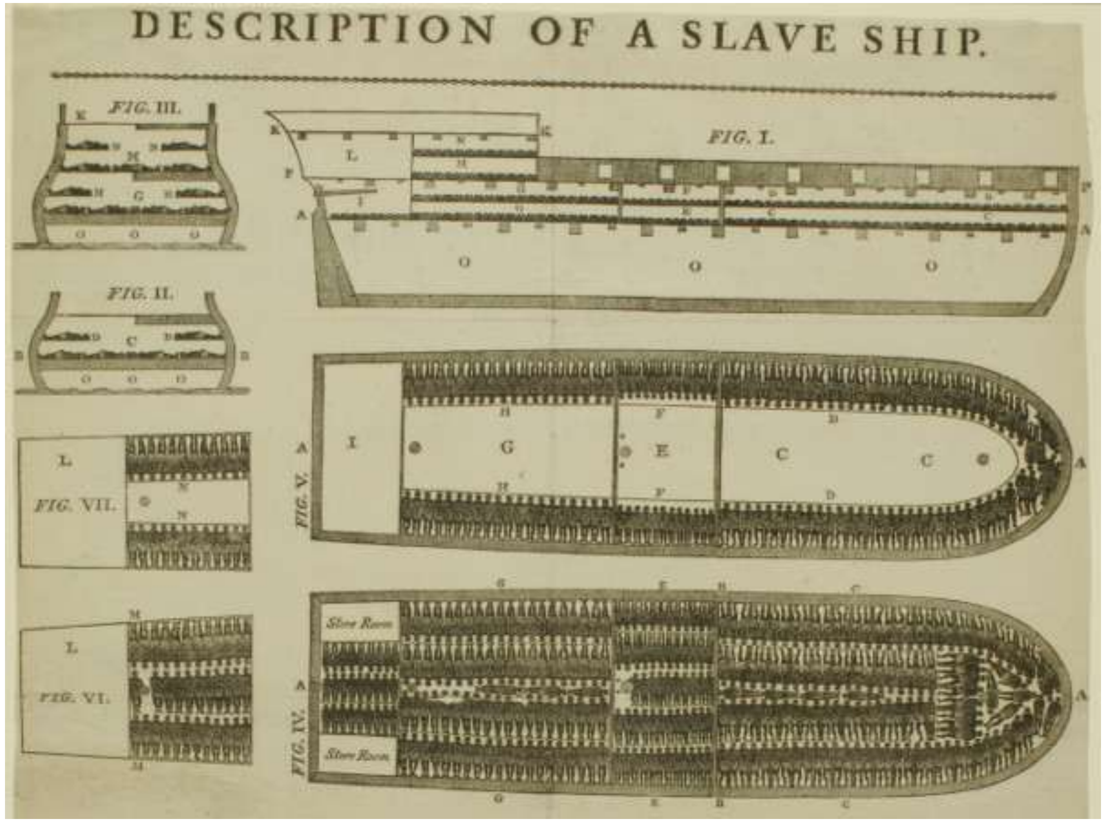
Visualization: A Way of Viewing the World



Ptolemaic map of the world by Bartolomeu Velho (1568 AD)

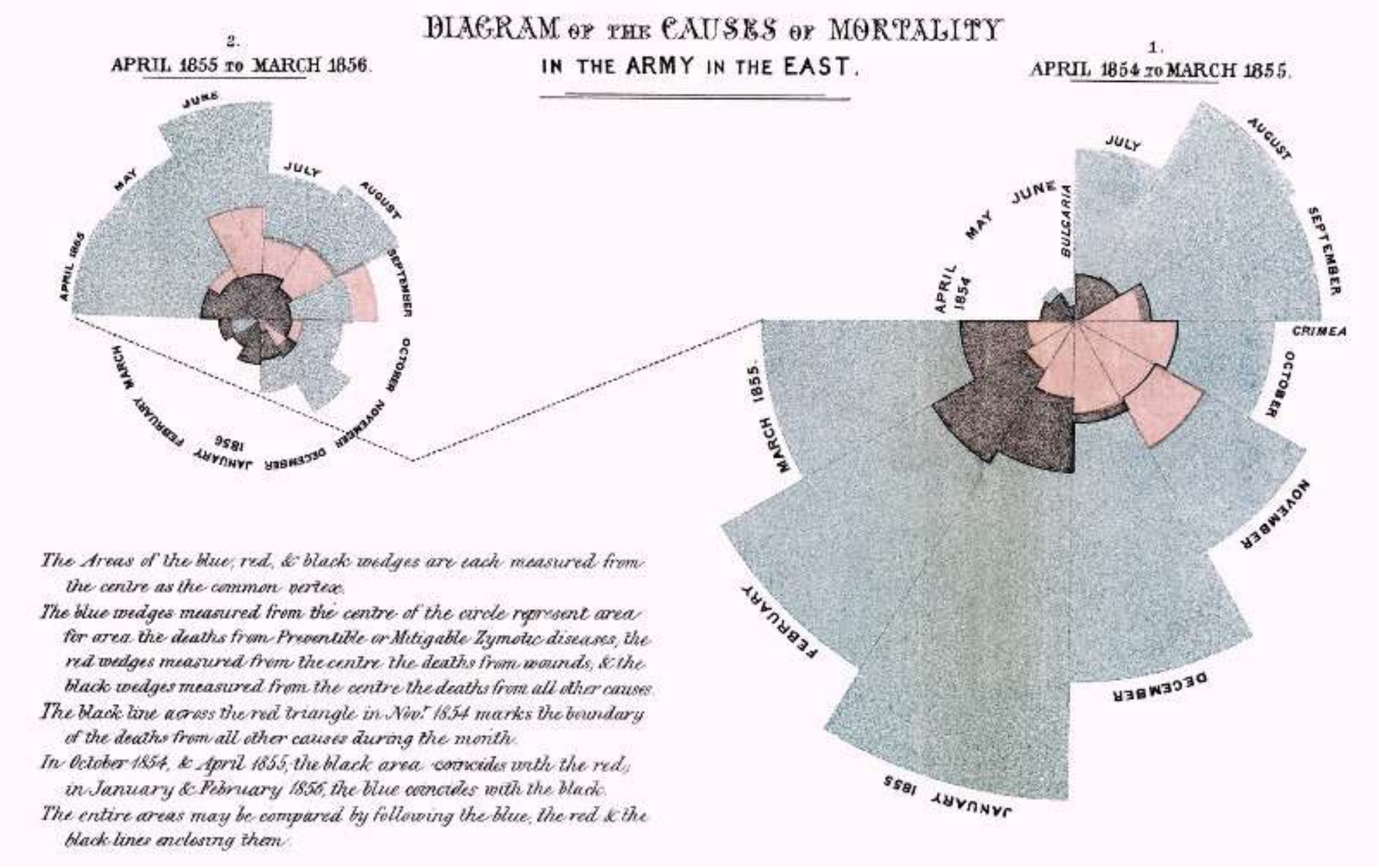
Motivation of Political Action

- Visualization of conditions on a slave ship was widely used by the antislavery movement in London 1789



Convincing the Parliament

- Nurse Florence Nightingale produced these plots to convince the British parliament in the 1850s of the importance of sanitation



Saving Lives Through Visualization

- In 1854, visualizing the locations of deaths (horizontal lines) helped John Snow deduce that a cholera epidemic was caused by a polluted water pump

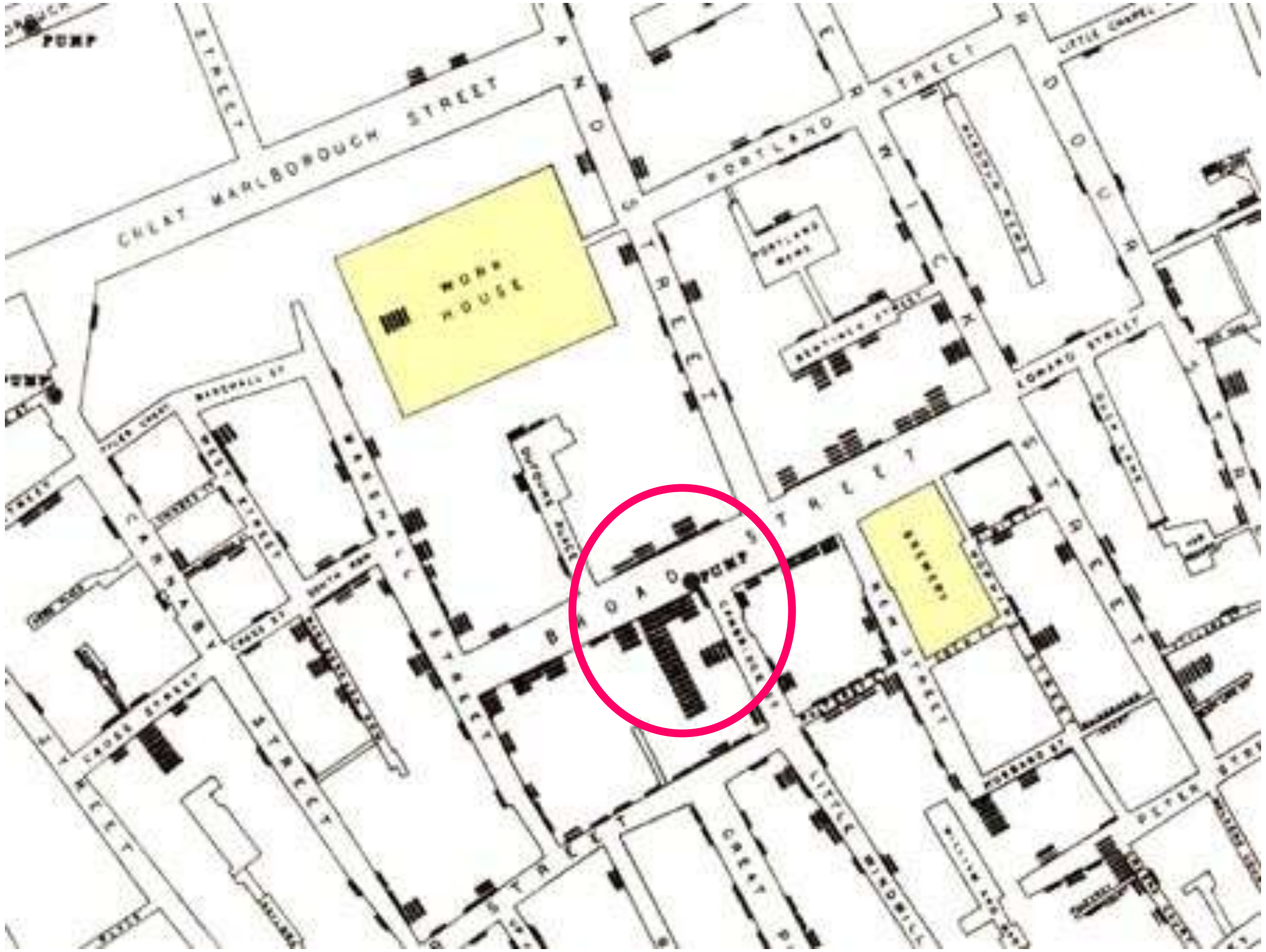
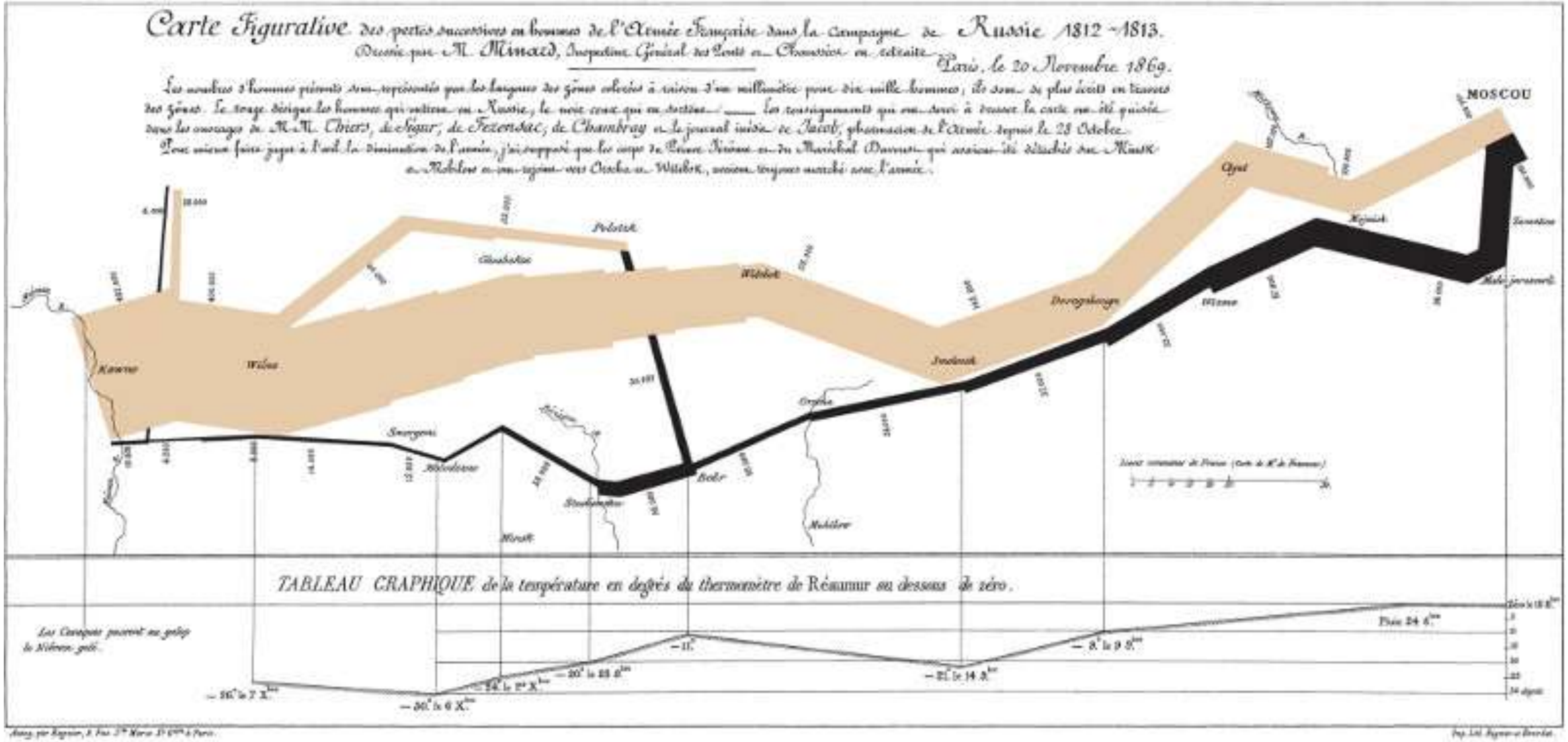
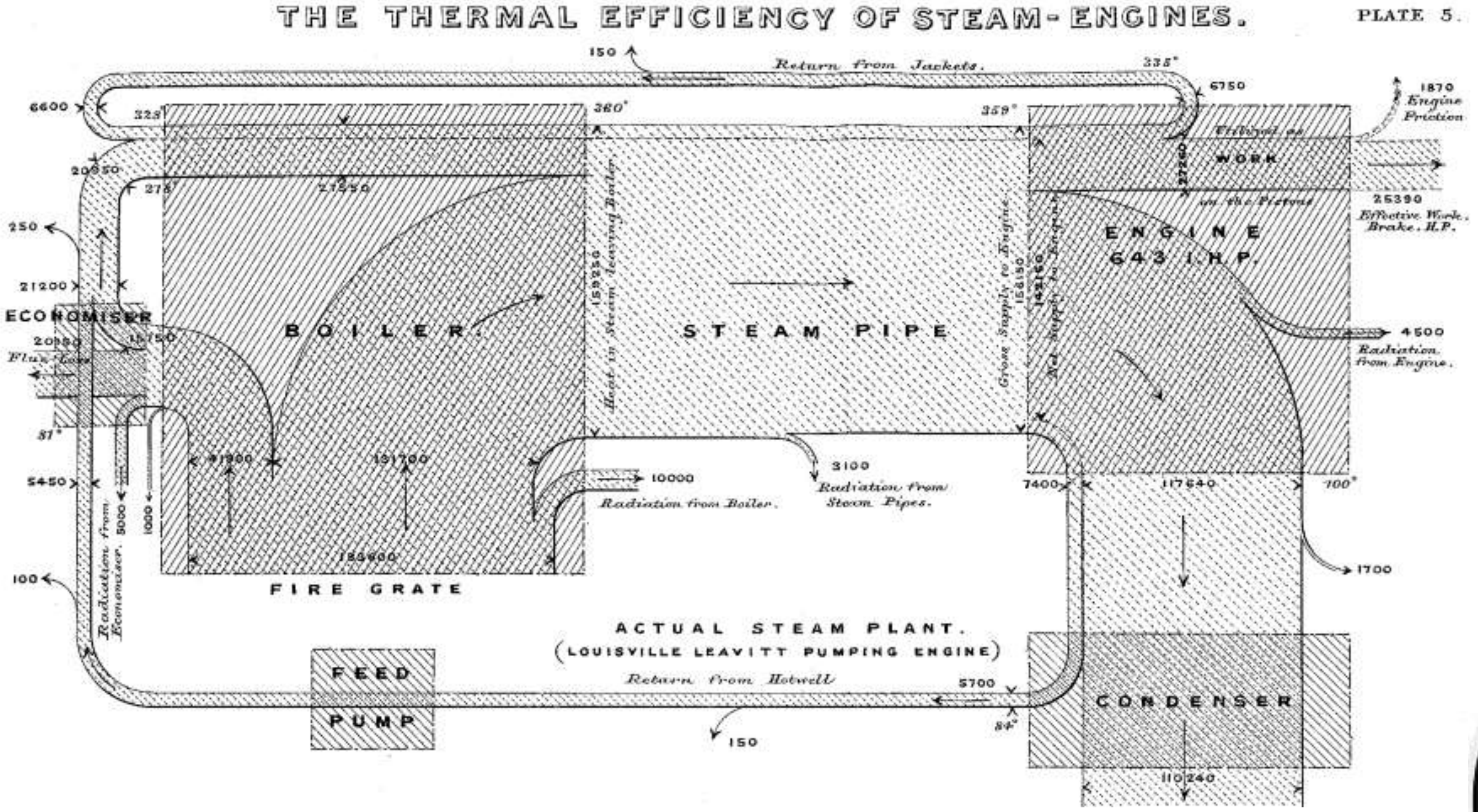


Illustration of Historical Events



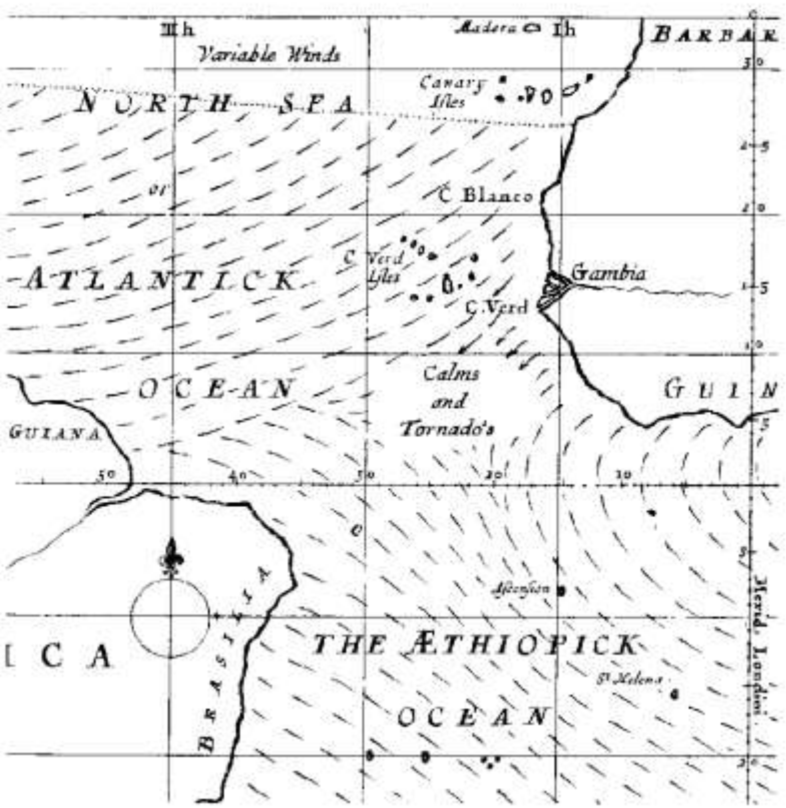
Napoleon‘s march on Russia (1812/13)
Cartography by Charles Joseph Minard (1861)

Visualization in Engineering – Sankey Diagrams

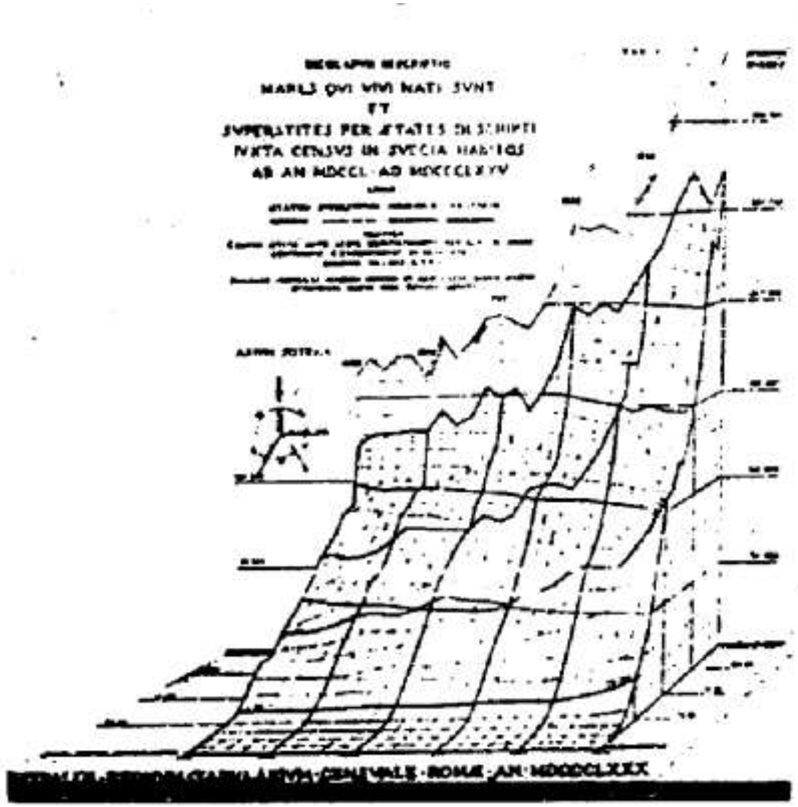


Energy Flow Diagram in a Steam Engine (1898)
Made by Capt. M.H.P.R. Sankey (1853-1925)

Height and Vector Field Visualizations



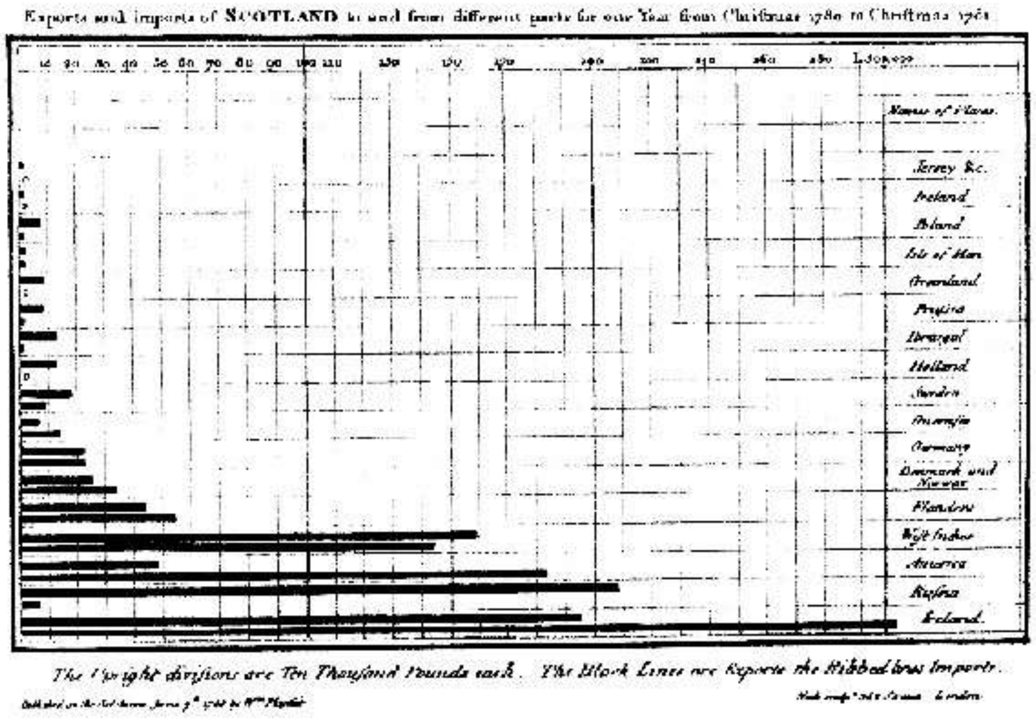
Vector visualization (1686)



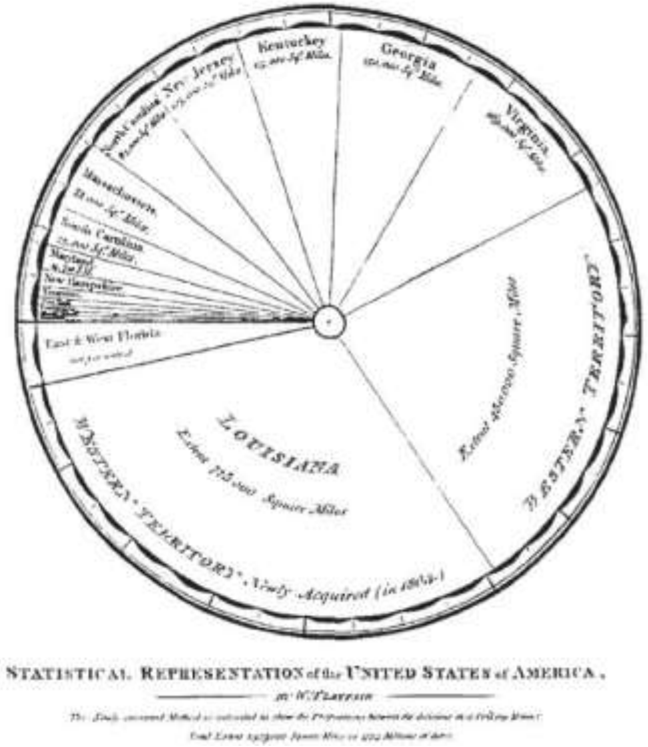
Height field (1879)

Statistical Visualization

- **William Playfair** (1759-1823)
  - Introduced statistical charts that today are available in any spreadsheet program



Bar Chart



Pie Chart

Viewing Outer Space

- Hand-colored visualization of the first closeup picture of Mars, taken by Mariner 4 in 1965

Visualization as a Scientific Community

A scientific community that specializes in visualization has existed for 35 years:

1990 IEEE Visualization conference

1995 IEEE Trans. on Vis. and Comp. Graphics

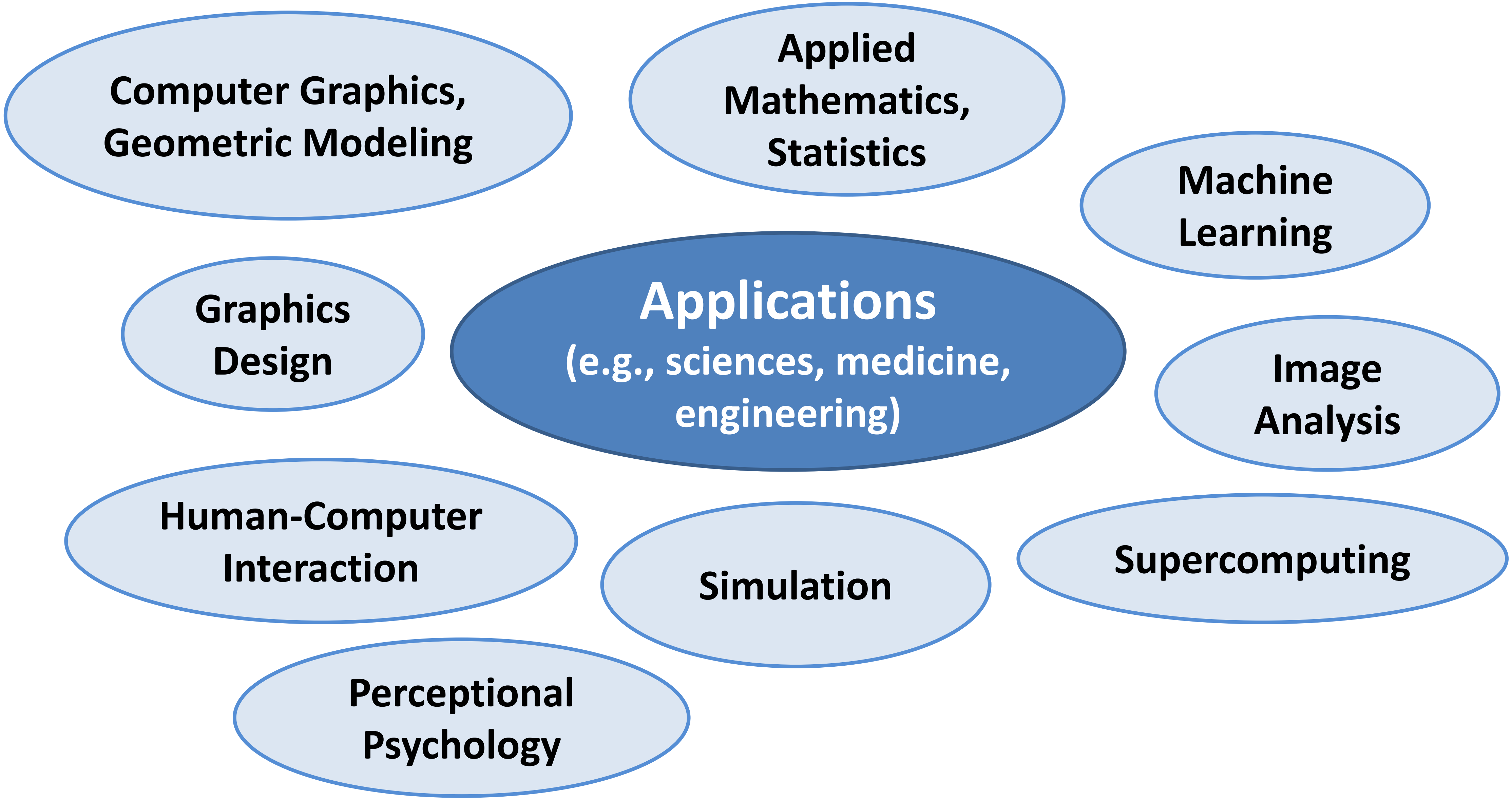
1999 IEEE/EG Symposium on Visualization

2008 Pacific Symposium on Visualization

2012 EuroVis achieves conference status

2021 IEEE VIS fully unifies former SciVis, InfoVis, VAST

How Visualization Relates to Other Fields



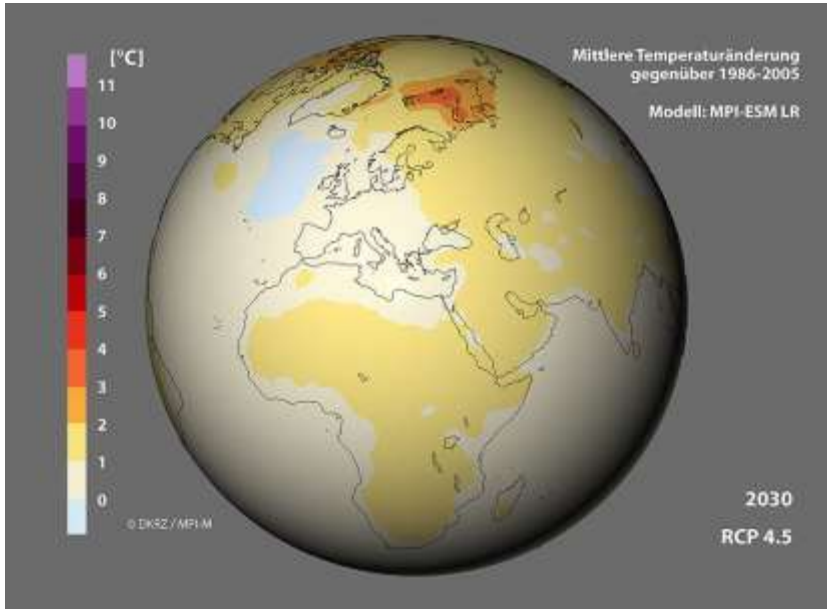
1.5 Examples of Visualization

Weather Maps

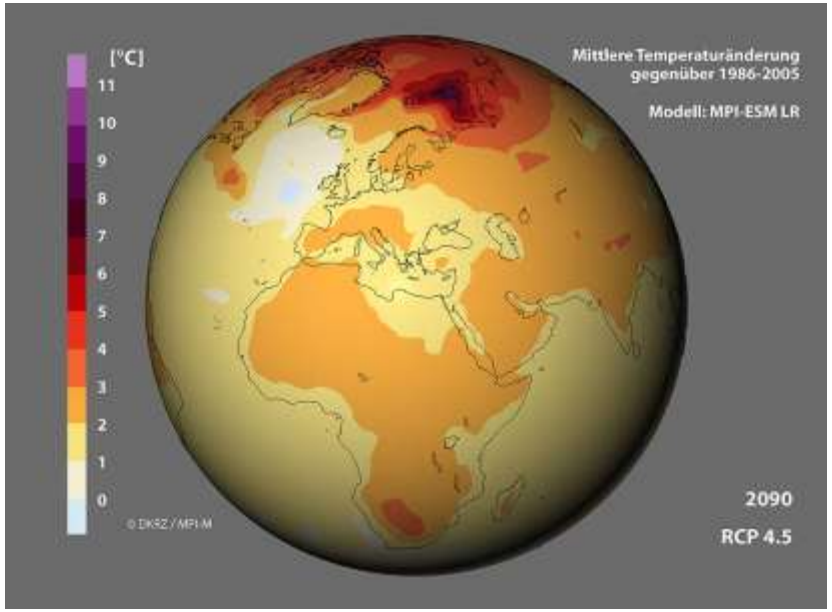
- Source: www.wetteronline.de

Climate Simulations

- Source: German Climate Research Center

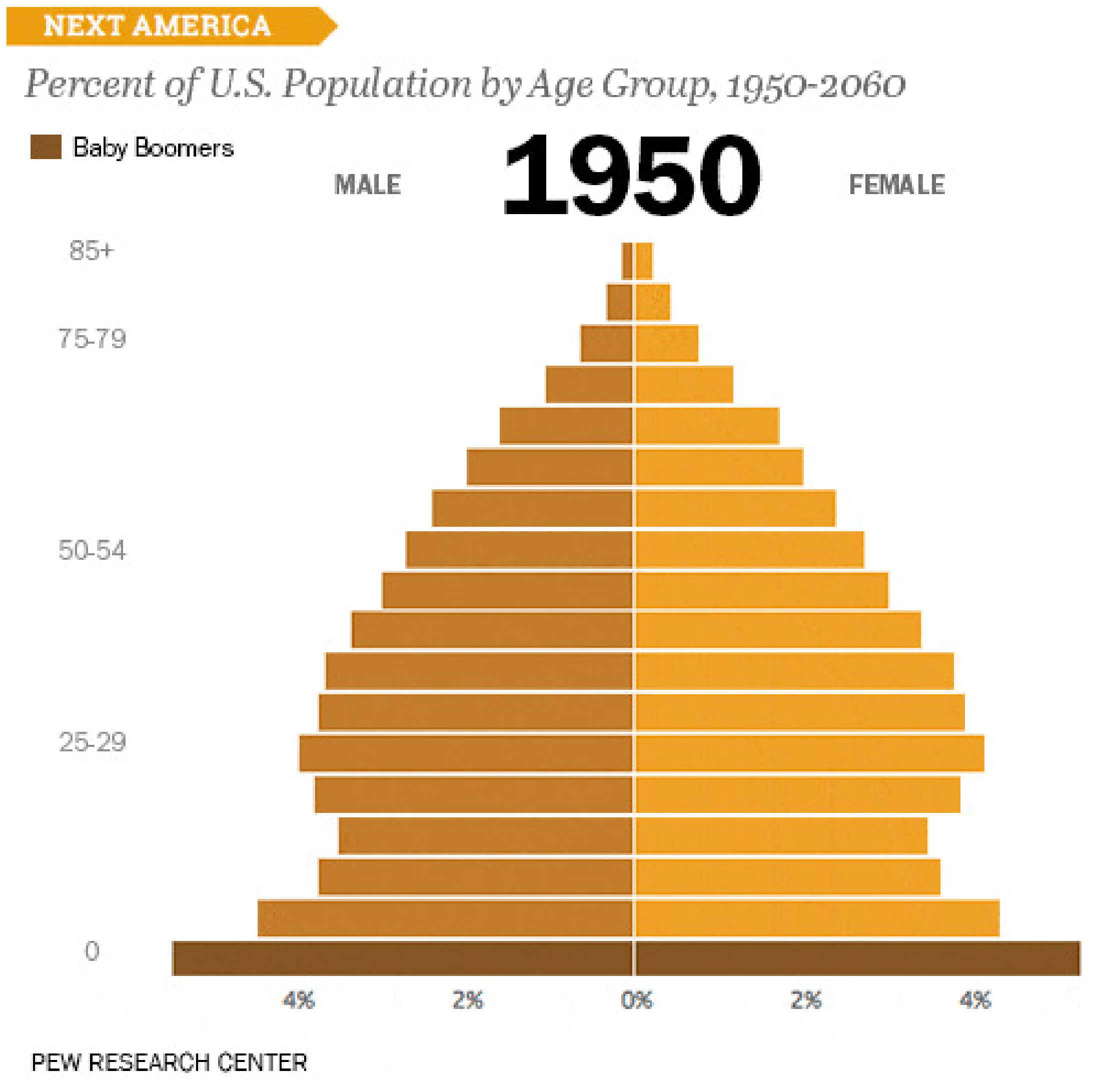


2030



2090

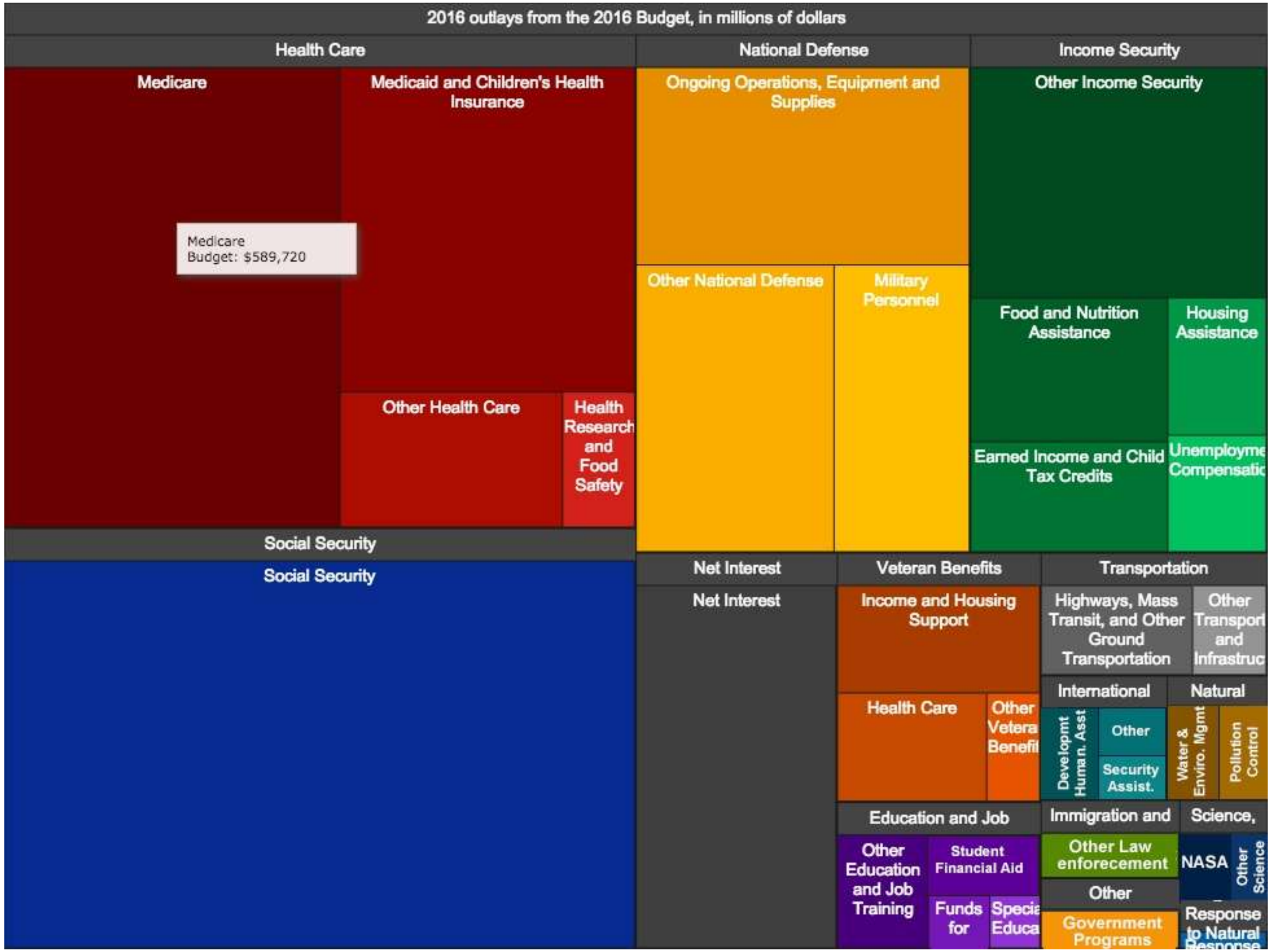
Demographic Change



Source: https://www.pewresearch.org/social-trends/2014/04/10/next-america/

Politics

- In 2016, the Obama administration used an interactive treemap to communicate the Federal budget to taxpayers



Source: https://obamawhitehouse.archives.gov/interactive-budget

Visualization for Educating the Public

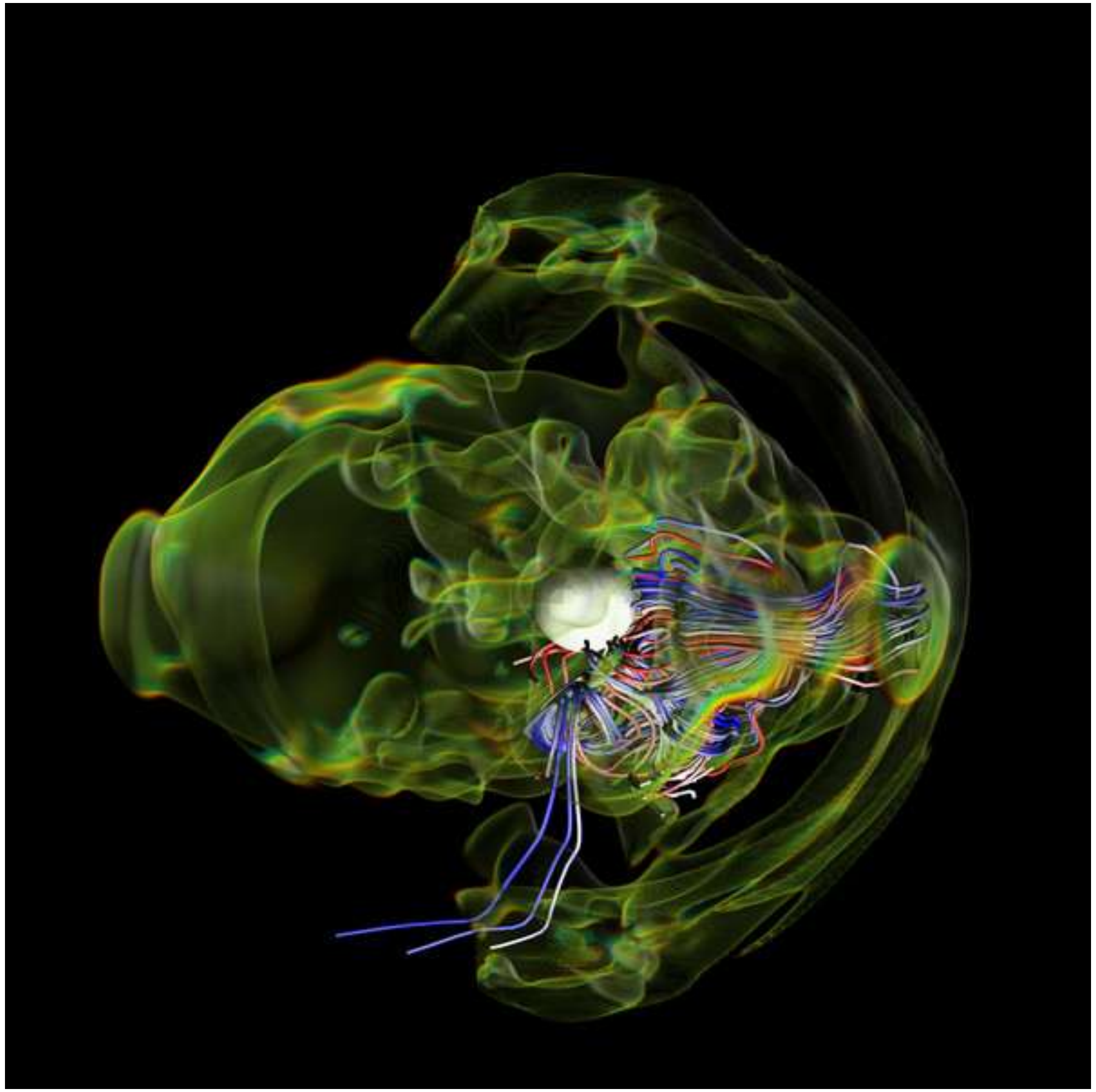
- The **Gapminder** project tries to correct widespread misconceptions about the development of different countries

https://www.gapminder.org/

https://www.youtube.com/watch?v=jbkSRLYSojo

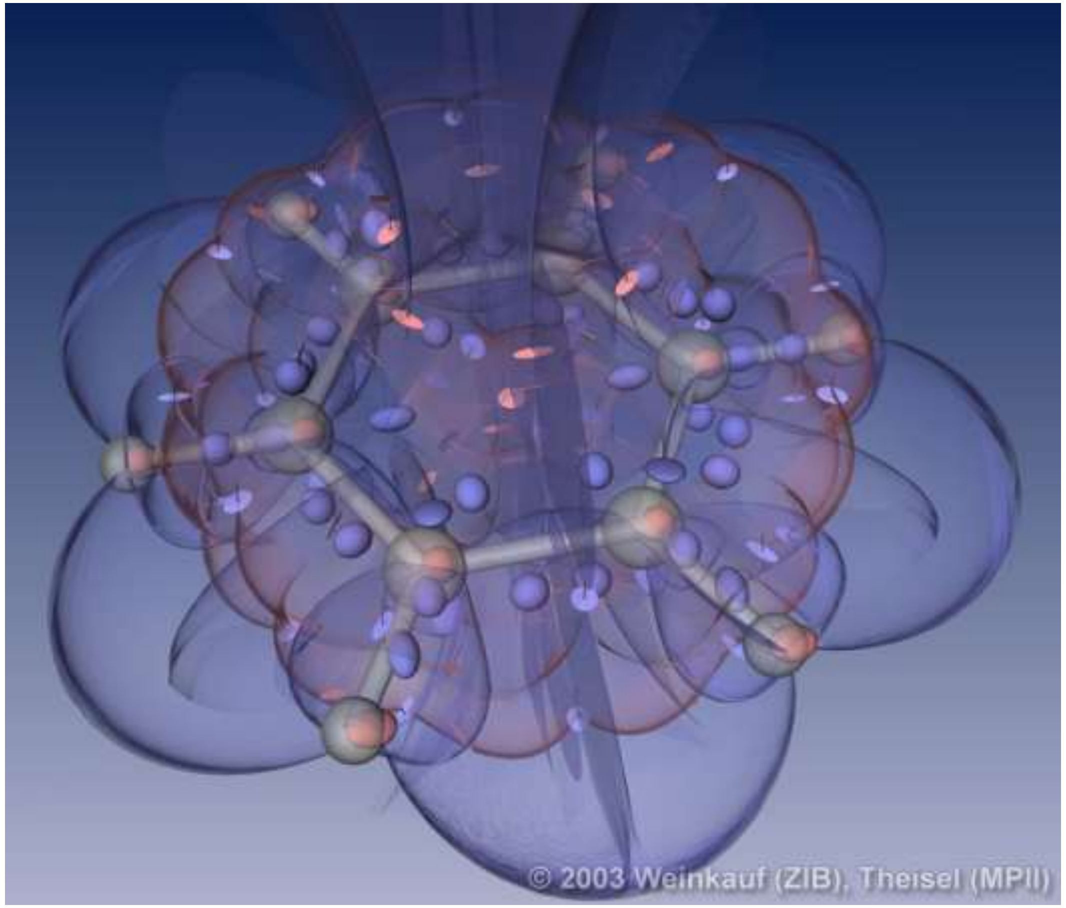
Astrophysics

Simulation of a supernova by Eirik Endeve, Christian Cardall, and Reuben Budiardja; visualized by Dave Pugmire



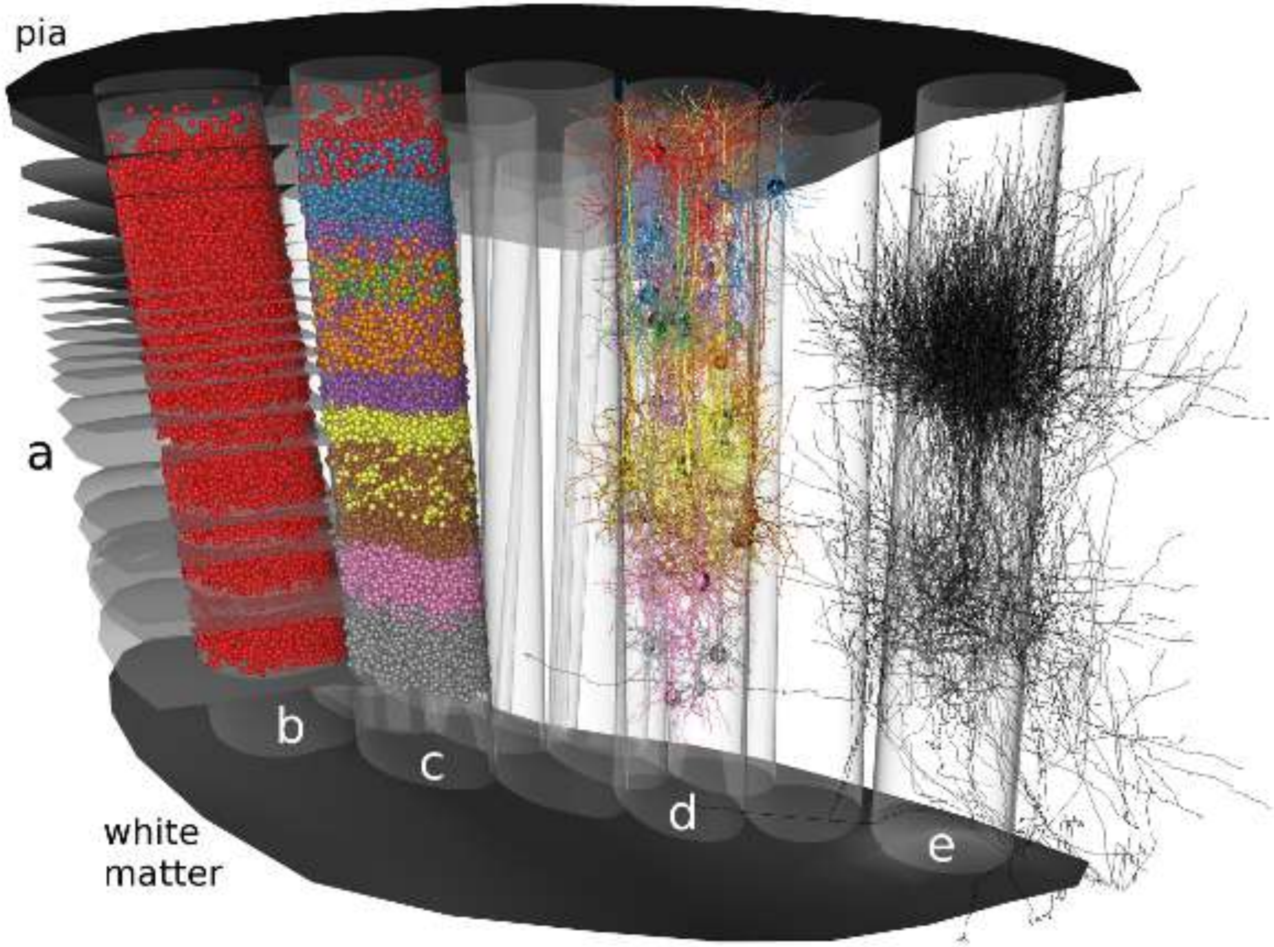
Chemistry

- Source: Tino Weinkauf and Holger Theisel



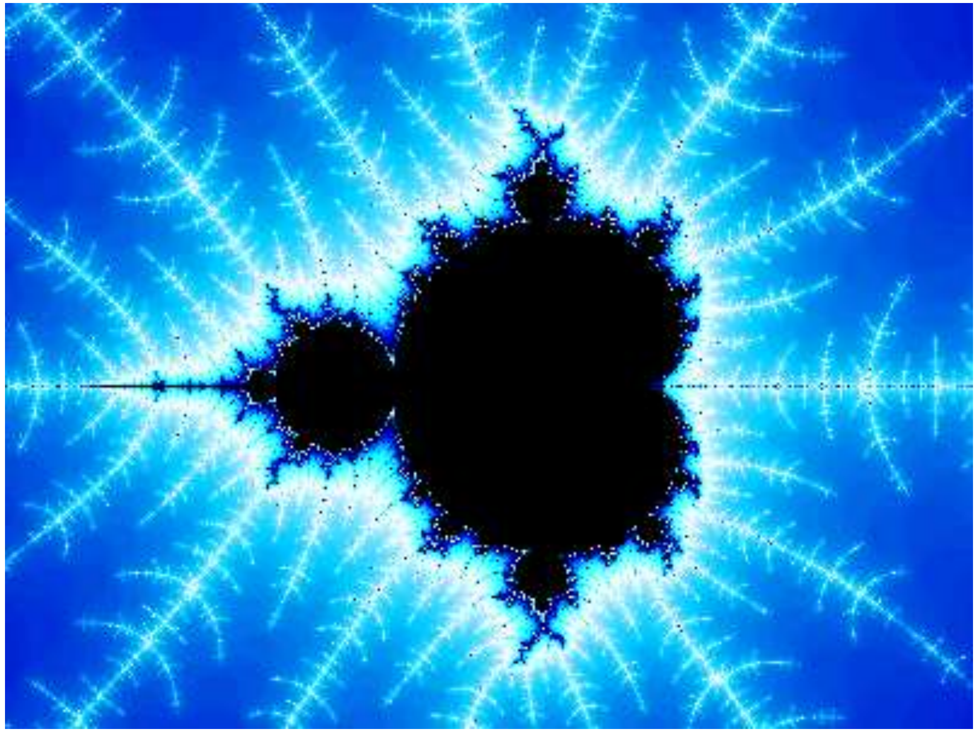
Neurobiology

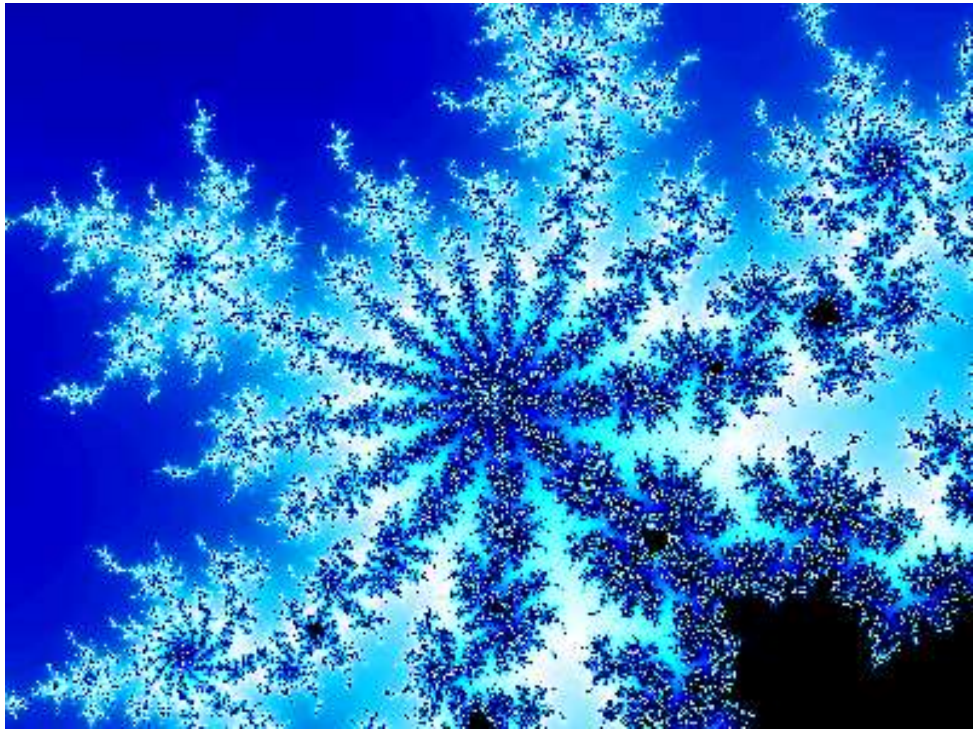
- Source: Zuse Institute Berlin



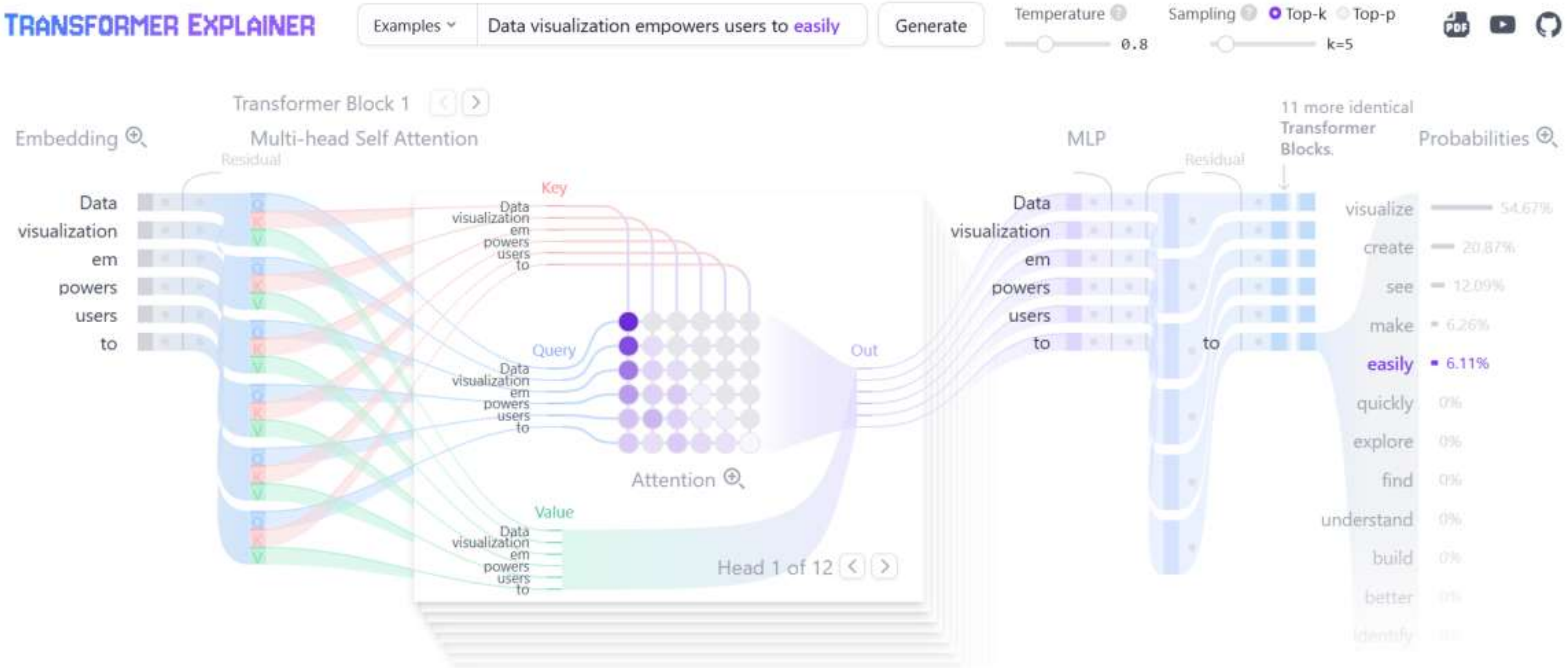
Mathematics

- Example: Visualization of Fractals
- Source: www.coolmath.com





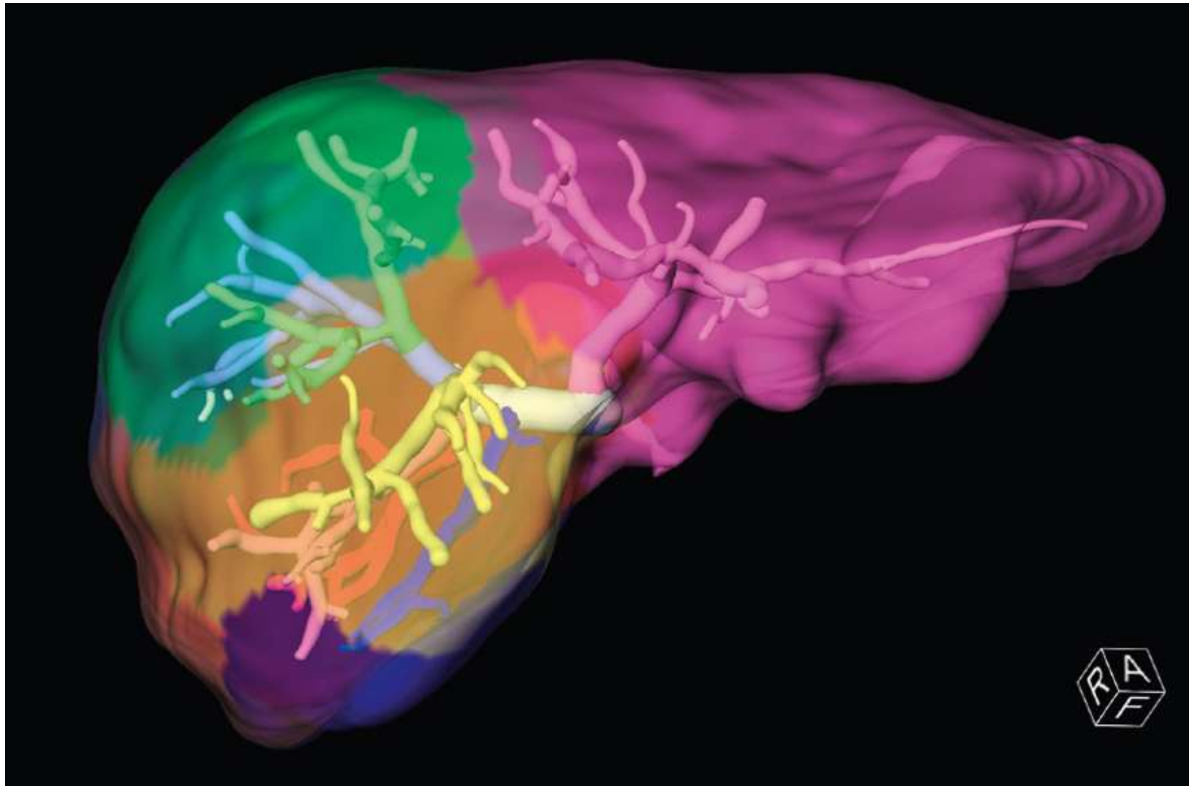
Education



https://poloclub.github.io/transformer-explainer/

Medicine

- Example: Liver Surgery (Source: Siemens)
- Example: 3D Ultrasound (Source: 3DBabyVu)



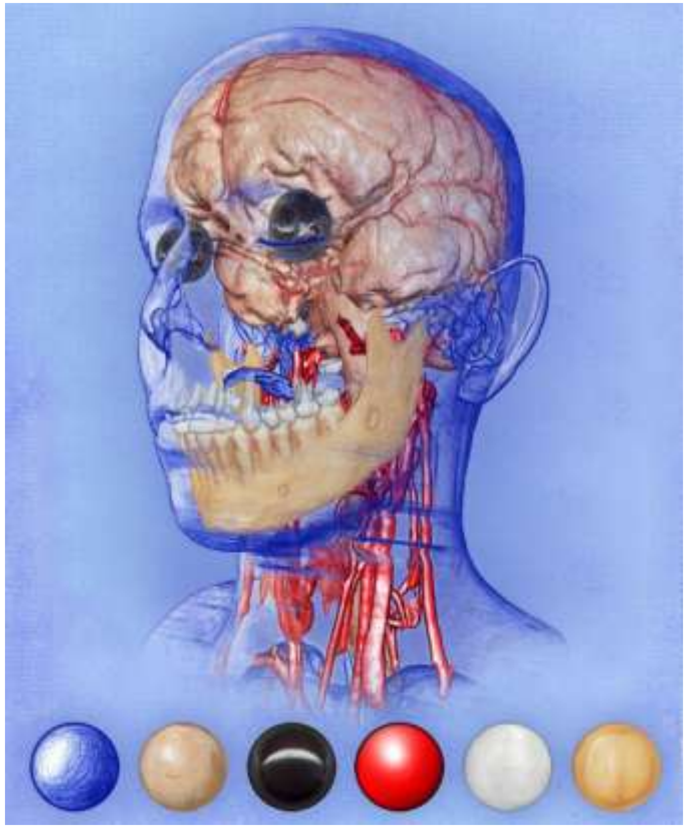
German Future Prize 2017

- **Cinematic rendering** of volume datasets won the „Deutscher Zukunftspreis“

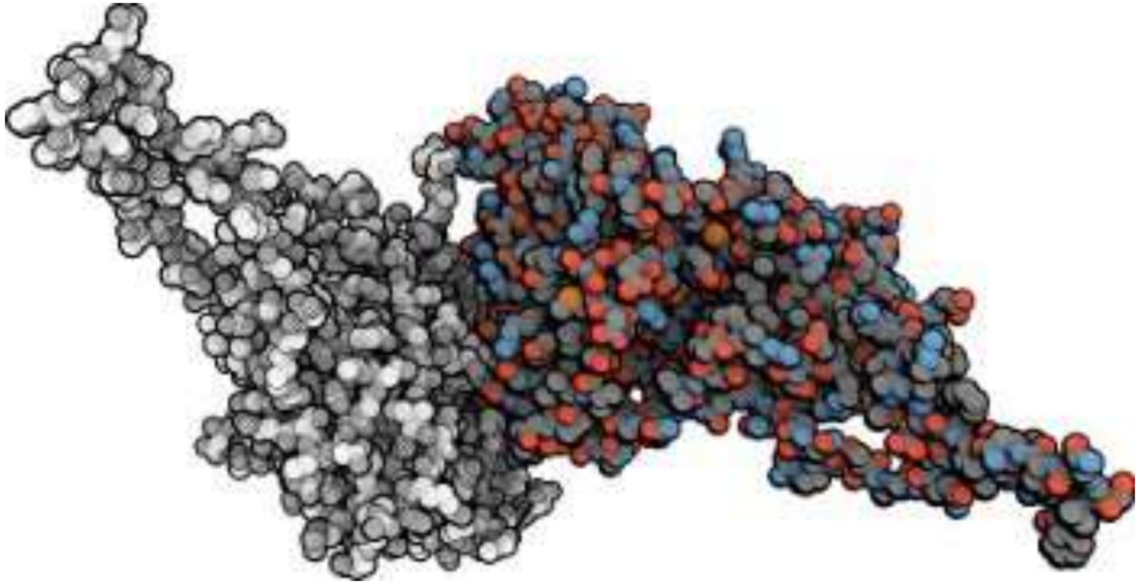
Image Source: Siemens Healthineers, Copyright: Hospital do Coracao, Sao Paulo, Brasil

Illustrative Visualization

- Illustrative Visualization renders real data in a style similar to manual textbook illustrations

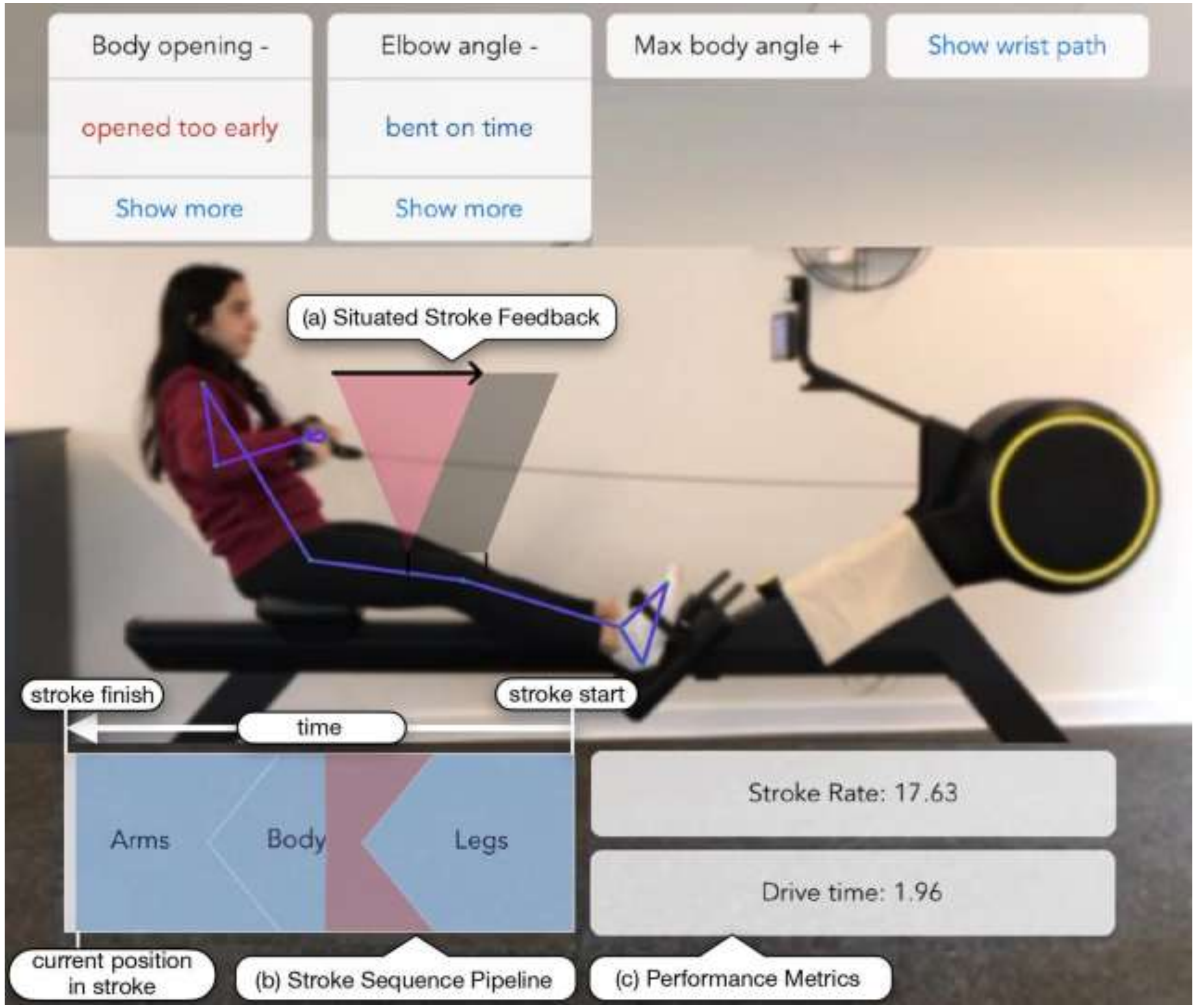


[Bruckner / Gröller 2007]



[van der Zwan et al. 2011]

Visualization in Sports



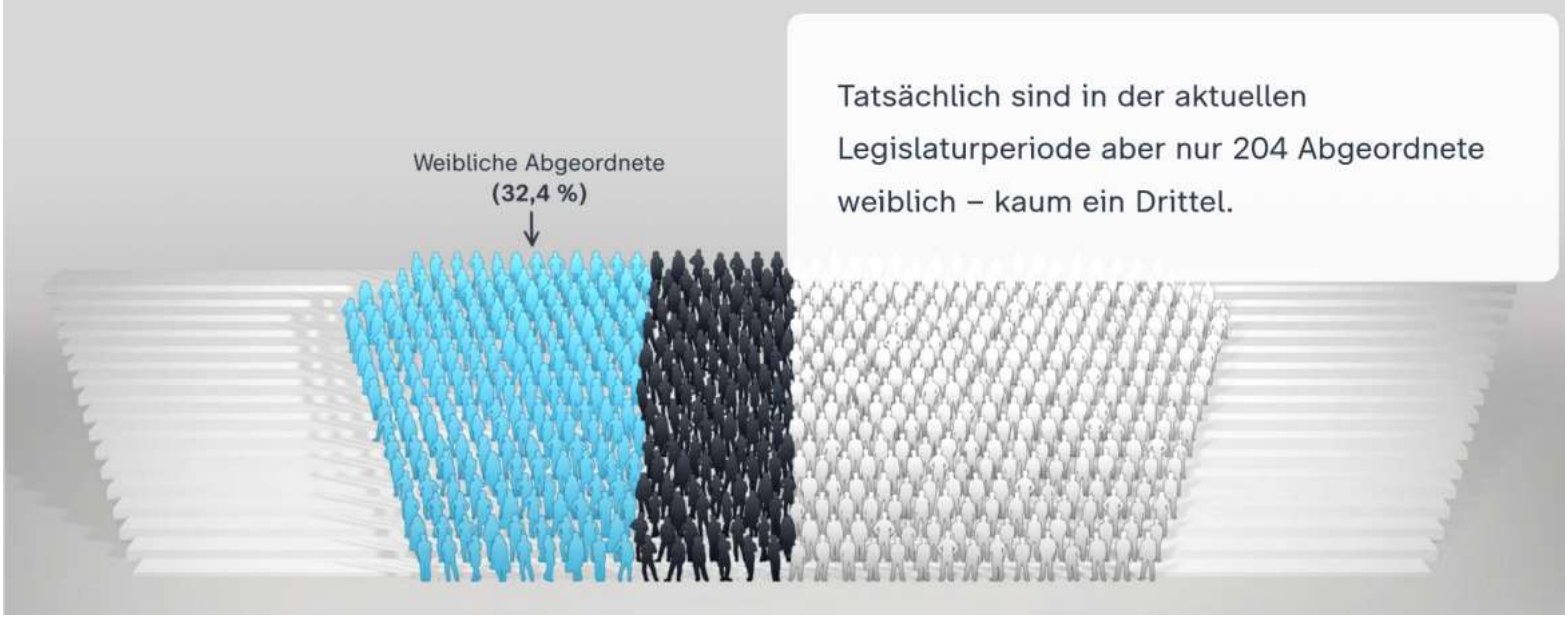
Iannucci et al.: *ARrow:* A Real-Time AR Rowing Coach. EuroVis SP 2023

Visualization in Journalism

Source: New York Times, February 26, 2022

Source: New York Times, March 21, 2022

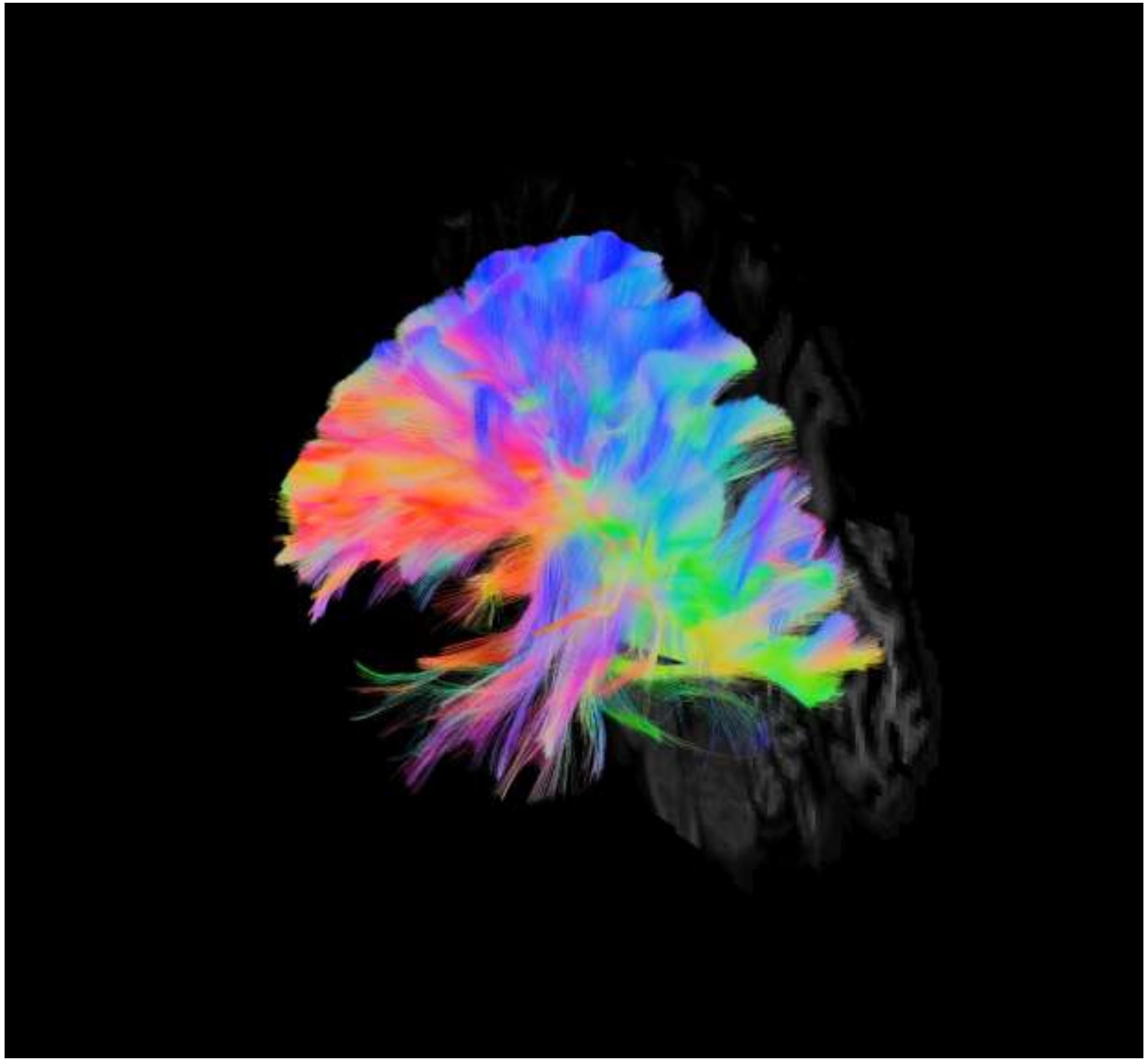
Data Driven Journalism



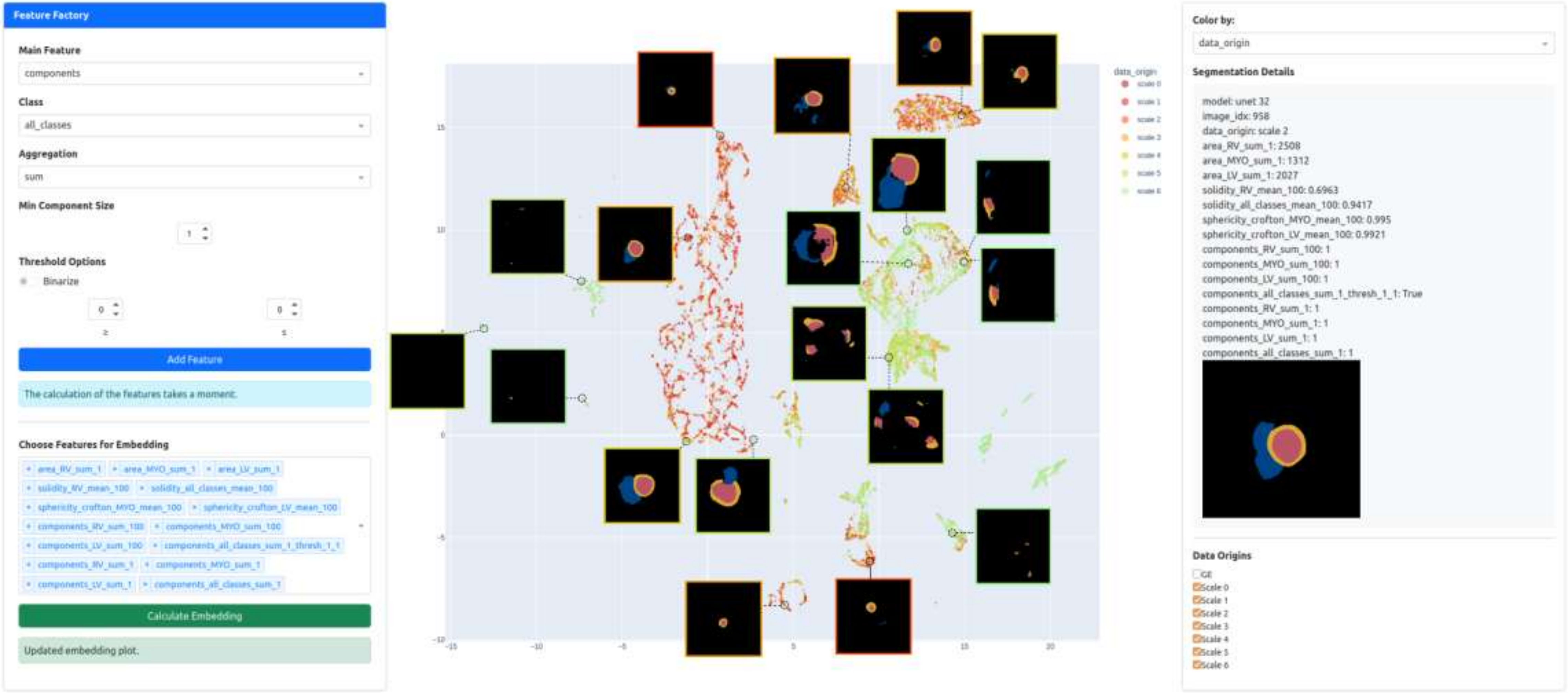
Source: ZDF, March 23, 2025

Vis@Bonn: Neuroimaging

Unscented Kalman Filter based multi-fiber tractography of *corpus callosum,* seeded at mid-sagittal plane
[Gruen et al. 2023]

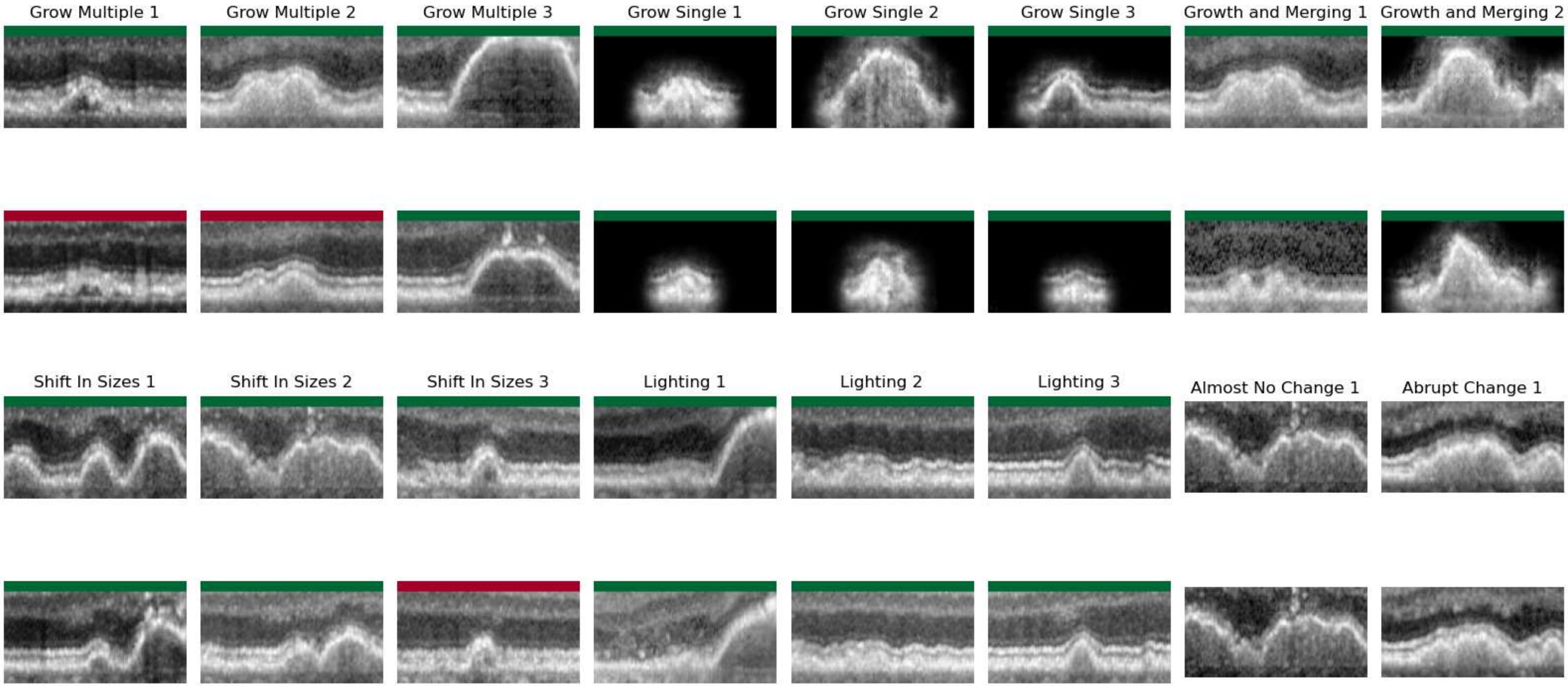


Vis@Bonn: Deep Learning based Image Analysis



[Mikliss/Schultz, EuroVis 2026]

Vis@Bonn: Generative AI for Visualization



[Muth et al., VCBM SP 2024]

Summary: Introduction to Visualization

- **Visualization** makes images out of data
  - Application-centered and highly interdisciplinary branch of computer science
  - *Goals:* Faster insight, clearer explanations, computational steering
  - Visualization has a long history
  - More recently, making sense of large and complex data, and clearly presenting your findings, has become a key qualification
    - Needed in science, engineering, medicine, business etc.
  - Visualization can also be creative and fun